On the Underlying Long Vowels in Contemporary Standard European Portuguese

Shinya Makino
itakanro@g.ecc.u-tokyo.ac.jp

Abstract:

Traditionally, Portuguese linguistics has acknowledged phonetic differences in vowel length conditioned by stress in Contemporary Standard European Portuguese (EP), yet has rarely addressed the possibility that underlying vowel length oppositions — not directly inherited from but analogous to those attested in Classical Latin and established through phonological developments in the 14th–15th centuries — may persist in the Contemporary language. In contrast, Makino (2006[1], 2006[2], 2007[2]) argued for the persistence of an underlying opposition in vowel length even in Contemporary EP, seeking to account for a range of phonological phenomena: the characteristic vowel alternations in open syllables preceding prepalatal and palatal consonants [ʃ ʒ ʎ ɲ]; the phonetic outcomes of vowel coalescence at the level of phrasal phonology; non-reduction in radical vowels of derived nouns and adjectives formed by suffixation to verbal radicals lacking intervening thematic vowels; and the absence of vowel reduction in unstressed syllables of derived words with negative implicatures. This theoretical study revisits this issue and offers a novel account of two phenomena: the absence of vowel quality reduction in nasal vowels in unstressed syllables, and the opposition [ɐ́ː] ≈ [áː] in the verb endings *-amos* and *-ámos* in the first-person plural of the present and preterite indicatives. The analysis is developed within the framework of Stratal Optimality Theory, and employs a revised view of vowel feature geometry in which the traditional [±ATR] feature is reinterpreted as the terminal direction of a multistage tongue-height vector. Without introducing new phonetic data, except for some instrumental measurements that provide acoustic evidence for the underlying length distinction, this study attempts a reinterpretation of patterns previously regarded as difficult to analyse within a unified framework.

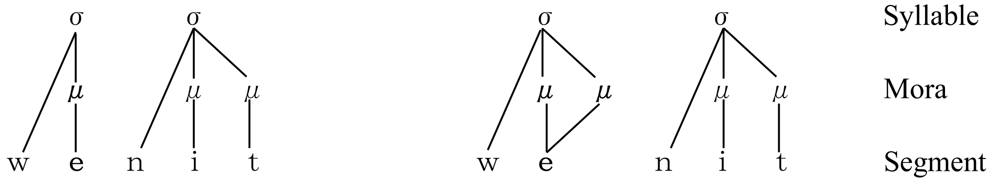
1. Introduction

In Contemporary Standard European Portuguese (henceforth EP), phonetic alternations in vowel length conditioned by stress can be observed — for example, [ɐ́ː]*ma* "he/she loves" versus [ɐ]*mamos* "we love" from the verb *amar* "to love". However, as far as we are aware, little to no attention has been paid to the possibility that Contemporary EP, like Classical Latin, may exhibit a phonological — that is, underlying — opposition in vowel length, as in *vĕnit* "he/she comes" versus *vēnit* "he/she came" from the verb *venīre* "to come". When one considers not only the stress-conditioned alternations in vowel length, but also the presence or absence of vowel quality alternations under the same conditioning, along with the etymological backgrounds of the relevant forms, it seems reasonable to hypothesise that a phonological opposition between short and long vowels — established by the year 1500 — may still persist at the underlying level in contemporary EP. Moreover, positing such an opposition makes it possible to offer a more unified analysis of phonetic patterns that might otherwise appear unrelated. Accordingly, this study does not aim to introduce new phonetic data, but rather constitutes a theoretical reinterpretation of well-known facts from an entirely different perspective — employing three fundamental conceptual devices: **the concept of the mora and the postulation of underlying long vowels, and the phonetic implementation of stress**.

2. Phonological vowel length

In this section, we adopt the perspective that distinguishes between the segmental tier — which encodes phonemic quality — and the moraic tier — which encodes phonological quantity — in representing vowel length. Classical Latin provides a canonical case of an underlying vowel length opposition, as in the minimal pair *vĕnit* [ˈvɛnɪt] "he/she comes" vs. *vēnit* [ˈveːnɪt] "he/she came". Their respective phonological representations may be schematised as in (1). (In the representations below, the final consonant /t/ is hypothetically treated as moraic; however, since the present analysis is concerned primarily with vowel quantity, the moraicity of /t/ is not crucial to the argument.)

(1) [ˈw**ɛ**nɪt] [ˈw**eː**nɪt]



The two forms are identical in that the segmental tier consists of the same sequence of segments — /w/, /e/, /n/, /i/, /t/ — in that order. The difference lies in whether the vocalic segment /e/ is linked to a single mora (μ) or to two moras. A mora is a unit of phonological weight and serves as a constituent of the syllable (σ) determining syllable weight — light vs. heavy. From the viewpoint of segment-to-mora association, a vowel segment (V) linked to a single μ yields a short vowel /$\overset{\mu}{\mathrm{V}}$/ — e.g. /w$\overset{\mu}{\mathrm{e}}$nit/ [ˈwɛnɪt] — whereas one linked to two moras corresponds to a long vowel /$\overset{\mu\mu}{\mathrm{V}}$/ — e.g. /w$\overset{\mu\mu}{\mathrm{e}}$nit/ [ˈweːnɪt]. As shown in (1), onset consonants are not associated with any mora and therefore do not contribute to phonological weight.

3. Underlying moraic specification of vowel segments in Contemporary EP

In this section, we examine how vowel segments in Contemporary EP are associated with moras, from the theoretical perspective outlined in Section 2. A key piece of evidence for this investigation is the pattern of phonological alternation in 3.1.

3.1. Vowel alternations conditioned by stress

In Contemporary EP, vowels typically exhibit phonetic alternations in both **quality** and **quantity**, as illustrated in (2), depending on whether they occur in stressed or unstressed syllable — for simplicity, the discussion in this section is restricted to open syllables — within the same morpheme. Teyssier (1984: 20) refers specifically to the alternation in vowel quality as "réduction". In certain morphemes, by contrast, only alternations in **quantity** are observed, without accompanying changes in quality, as shown in (3)[1]. (Note: the symbol ≈ indicates alternation, m.pt. stands for Middle Portuguese, < denotes a sound change, and ▭ marks a borrowing.)

(2)

[ɛ́ː] ≈ [ɨ]	*pr*[**ɛ́**ː]*ga* ≈ *pr*[**ɨ**]*gar* (< m.pt. *pr**e**gar* < lat. *plicāre*) from *pregar* "to nail"
[áː] ≈ [ɐ]	[**á**ː]*ta* ≈ [**ɐ**]*tar* (< m.pt. ***a**tar* < lat. *aptāre*) from *atar* "to tie"
[ɔ́ː] ≈ [u]	*m*[**ɔ́**ː]*ra* ≈ *m*[**u**]*rar* (< m.pt. *m**o**rar* < lat. *morārī*) from *morar* "to live"

[1] Makino (2006³) demonstrated acoustically that there is a statistically significant durational difference between stressed and unstressed syllables: the former are realized, on average, with approximately twice the duration of the latter. Since no substantial durational difference was observed in the consonantal segments, this difference in syllable duration is attributable primarily to vowel length.

(3)	**[έ:] ≈ [ɛ]**	*pr*[**έ**:]*ga* ≈ *pr*[**ɛ**]*gar* (< m.pt. *pr**ee**gar* < lat. *praedicāre*) from *pregar* "to preach" *refl*[**έ**:]*te* ≈ *refl*[**ɛ**]*tir* (< m.pt. *refl**ect**ir* □ lat. *reflectere*) from *refletir* "to reflect"
	[á:] ≈ [a]	*g*[**á**:]*nha* ≈ *g*[**a**]*nhar* (< m.pt. *g**aa**nhar* < lat. **guadaniāre*) from *ganhar* "to gain" [**á**:]*to* ≈ [**a**]*tor* (< m.pt. ***acto/actor*** □ lat. *āctus/āctor*) *ato* "act" and *ator* "actor"
	[ɔ́:] ≈ [ɔ]	*c*[**ɔ́**:]*ra* ≈ *c*[**ɔ**]*rar* (< m.pt. *c**oo**rar* < lat. *colōrāre*) from *corar* "to brush" *ad*[**ɔ́**:]*ta* ≈ *ad*[**ɔ**]*tar* (< m.pt. *ad**op**tar* □ lat. *adoptāre*) from *adotar* "to adopt"

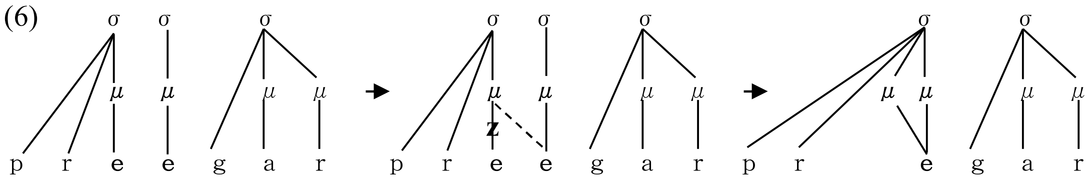
From an etymological perspective, the vowel involved in the alternation in (2) derives from a monophthong (e.g. *prega* ≈ *pregar*), whereas the vowel in (3) originates in many cases from one of the following sources (4) and (5):

(4) **a sequence of two monophthongs** — resulting in a long vowel via contraction (e.g. *pr**ee**ga* ≈ *pr**ee**gar*);

(5) **a monophthong followed by a coda consonant** — yielding a long vowel through compensatory lengthening triggered by consonant deletion (e.g. *refl**ec**te* ≈ *refl**ec**tir*).

Assuming the two-tiered representation proposed in Section 2 — which distinguishes between the segmental tier and moraic tier — the historical changes that the m.pt. *preegar* is likely to have undergone by "por volta de 1500" (cf. Teyssier 1980: 43) may be schematically represented as in (6):

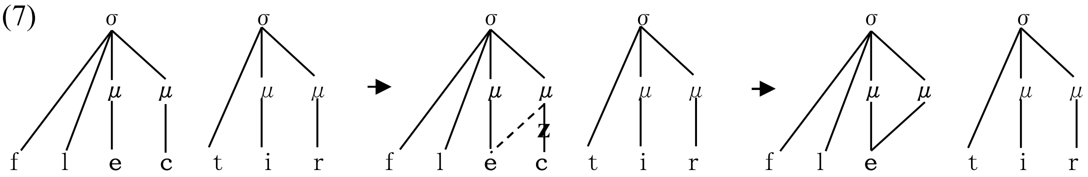
(6)



In the initial stage of (6), the two adjecent pretonic vowels *e* were each linked to a single mora, forming two distinct syllables: *pr$\overset{\mu}{e}$-* and *-$\overset{\mu}{e}$-*. However, as a result of "contracção" — a vowel coalescence process that occurred during the 14th to 15th centuries (cf. Teyssier 1980: 40-43), likely triggered by the Obligatory Contour Principle (OCP) — one of the two *e*s was delinked from its associated mora. (It is not certain which of them was actually deleted; however, given that the second *e* was the radical vowel of the verb and stressable in rhizotonic forms, it might have had more stability than the first *e*.) That mora was then relinked to the other $\overset{\mu}{e}$, resulting in compensatory lengthening in the second stage. Simultaneously, resyllabification took place in the third stage, yielding a new syllable *pr$\overset{\mu\mu}{e}$-*, with an underlying long vowel /$\overset{\mu\mu}{ɛ}$/ as its nucleus.

In the case of *reflectir* in (3), a borrowing from Classical Latin, the historical development follows essentially the same pattern, as illustrated in (7). For reasons of space, the prefix *re-* is omitted in the representation.

(7)



In this case, it is assumed that the coda consonant *-c* was deleted due to the tendency of PE to avoid closed syllables (cf. Makino 2004[2]: 207) — indicated by the delinking of *-c*, represented by **z** on the association line — and that the mora previously linked to *-c* was associated to the preceding $\overset{\mu}{e}$, resulting in a underlyingly long vowel /$\overset{\mu\mu}{ɛ}$/ linked to two moras.

As Teyssier (1980: 42) notes: "No século XV, quando as contracções das vogais em hiato se completaram, essas vogais deviam ser *longas* e *abertas*, em oposição às pretónicas simples [ẹ] [ä] e [ọ], que eram *breves* e *fechadas*", which would be in IPA [e] [ɐ] and [o], respectively. Although the open-mid long vowel /$\overset{\mu\mu}{ɛ}$/ subsequently lost its phonetic length, it has retained its vowel quality to this day, as seen in *pr*[ɛ]*gar* and *refl*[ɛ]*tir*. On the other hand, the pretonic *e* in *pr$\overset{\mu}{e}$gar* in (2), which was etymologically a monophthong, did not undergo such historical changes and is assumed to have remained an underlying short vowel /$\overset{\mu}{ɛ}$/ linked to only one mora. After passing through the stage of [e] "*breve* e *fechada*" above in the 15th century, it subsequently underwent a phonetic change "no século XVIII, provavelmente depois de 1750" (Teyssier 1980: 62), and is today realised in the form *pr*[ɨ]*gar*.

This situation is assumed to hold not only for *e* /ɛ/, but also for *a* /a/ and *o* /ɔ/, and likewise for stressed vowels. Accordingly, between the alternations in vowel quality and quantity conditioned by the presence or absence of stress in the Contemporary language and their earlier states, a relationship of the type illustrated in (8) can be observed:

(8)

Middle EP (ca. 1500)		Contemporary EP
Underlying short vowels: /$\overset{\mu}{\acute{ɛ}}$ $\overset{\mu}{\acute{a}}$ $\overset{\mu}{\acute{ɔ}}$/ ≈ /$\overset{\mu}{ɛ}$ $\overset{\mu}{a}$ $\overset{\mu}{ɔ}$/	⇒	**[ɛ́ː áː ɔ́ː] ≈ [ɨ ɐ u]**
Underlying long vowels: /$\overset{\mu\mu}{\acute{ɛ}}$ $\overset{\mu\mu}{\acute{a}}$ $\overset{\mu\mu}{\acute{ɔ}}$/ ≈ /$\overset{\mu\mu}{ɛ}$ $\overset{\mu\mu}{a}$ $\overset{\mu\mu}{ɔ}$/		**[ɛ́ː áː ɔ́ː] ≈ [ɛ a ɔ]**

3.2. The opposition between short and long vowels at the underlying level and their phonetic realisation

What becomes clear from the vowel alternations in Contemporary EP illustrated in (8) is the following:

(9) The alternation in quantity — [ɛ́ː áː ɔ́ː] ≈ [ɨ ɐ u] and [ɛ́ː áː ɔ́ː] ≈ [ɛ a ɔ] — is determined by the presence or absence of stress: vowels are realised as bimoraic long vowels in stressed syllables — particularly in stressed open syllables — and as monomoraic short vowels in unstressed syllables.

(10) The alternation in quality — [ɛ́ː áː ɔ́ː] ≈ [ɨ ɐ u] — is likewise conditioned by stress, with vowel quality reduction occurring in unstressed syllables.

(11) Only those cases in which vowel quality reduction does not occur — such as those exemplified in (3) — are not synchronically predictable.

The alternation in quantity (9) and the alternation in quality (10) are both predictable from the presence or absence of stress, and therefore pertain to the grammar. By contrast, (11) is unpredictable and constitutes lexical information. Although previous studies in generative tradition have not made this point explicit, it might be inferred that both the vowels showing qualitative alternation — [ɛ́ː áː ɔ́ː] ≈ [ɨ ɐ u] — and those that do not — [ɛ́ː áː ɔ́ː] ≈ [ɛ a ɔ] — are underlyingly represented by the same vowel segments /ɛ a ɔ/, and that the absence of *réduction* in vowel quality is thus considered a lexical idiosyncrasy specific to the lexical item in question.

In contrast, based on the phonological principle that differences in surface alternation patterns in the same context may reflect differences in underlying representations, the present study proposes the hypothesis that **the underlying opposition between short and long vowels, established by the year 1500, has persisted into Contemporary EP**.

The reasoning behind this hypothesis is as follows. The alternations [áː] ≈ [ɐ] and [áː] ≈ [a] are asymmetrical in that, although it is not predictable whether [áː] alternates with [ɐ] or [a] in unstressed syllables, the converse alternation — namely, that [ɐ] and [a] alternate with [áː] in stressed syllables — is entirely predictable. Accordingly, if one sets aside preconceptions, it is the opposition between [ɐ] and [a] in unstressed syllables that should be considered underlying and, consequently, the changes [ɐ] → [áː] and [a] → [áː] indicate that the underlying opposition between [ɐ] and [a] is neutralised in stressed syllables. Nevertheless, since [ɐ] occurs only in unstressed syllables, it is implausible to assume that it is stored in the lexicon in that form. Therefore, this study assumes a full vowel /a/ as the underlying representation of [ɐ]. Under this interpretation, [áː] in stressed syllables is derived by lengthening the underlying /a/ as a phonetic implementation of stress assignment. This relationship can be summarised as in (12) using the notation μ for mora:

(12) /$\overset{\mu}{\text{á}}$/ ≈ /$\overset{\mu}{\text{a}}$/ → [áː] ≈ [ɐ]

On the other hand, if [ɐ] is assumed to be an unstressed allophone of /$\overset{\mu}{\text{a}}$/, unstressed [a] in the alternation [áː] ≈ [a] cannot be interpreted as a realisation of /$\overset{\mu}{\text{a}}$/, since, as argued above, [a] and [ɐ] constitute an underlying opposition. Given the alternation /$\overset{\mu}{\text{á}}$/ ≈ /$\overset{\mu}{\text{a}}$/ → [áː] ≈ [ɐ] in (12), it is evident that [ɐ] in unstressed syllables is realised with a loss of the distinct quality characterising [áː] in stressed syllables. If the alternation [áː] ≈ [a] runs in parallel with this, then what [a] in unstressed syllables is losing, compared to stressed [áː], is not its quality but its quantity — more precisely, one mora of length. It follows, then, that the underlying representation of [a] is /$\overset{\mu\mu}{\text{a}}$/. From this, the relationship (13) may be established:

(13) /$\overset{\mu\mu}{\text{á}}$/ ≈ /$\overset{\mu\mu}{\text{a}}$/ → [áː] ≈ [a]

The same line of reasoning can be applied to the other alternations as well. Therefore, **the alternation that is both qualitative and quantitative — [ɛ́ː áː ɔ́ː] ≈ [ɨ ɐ u] — which exhibits vowel quality reduction in unstressed syllables, corresponds to the phonetic realisation of the underlying short vowels /$\overset{\mu}{\text{ɛ}}$ $\overset{\mu}{\text{a}}$ $\overset{\mu}{\text{ɔ}}$/**, whereas **the uniquely quantative alternation [ɛ́ː áː ɔ́ː] ≈ [ɛ a ɔ], absent of such quality reduction, corresponds to the phonetic realisation of the underlying long vowels /$\overset{\mu\mu}{\text{ɛ}}$ $\overset{\mu\mu}{\text{a}}$ $\overset{\mu\mu}{\text{ɔ}}$/**.

Under this phonological interpretation of the relevant phonetic data in (8), **both underlying short vowels and underlying long vowels undergo**, accordingly, **some form of reduction in unstressed syllables**. Specifically, **the underlying long vowels**, which are bimoraic, **weaken by losing one mora** — that is, by **reducing their quantity** — while **the underlying short vowels**, which are monomoraic and thus have no mora to lose, **weaken by altering their feature values** — that is, by **reducing their quality**.

Moreover, the presence or absence of vowel quality reduction in (10) also becomes theoretically predictable as a reflex of the opposition between underlying short vowels and underlying long vowels. That is, in unstressed syllables, the short vowels /$\overset{\mu}{\text{ɛ}}$ $\overset{\mu}{\text{a}}$ $\overset{\mu}{\text{ɔ}}$/ — linked to a single mora in the underlying representation — undergo vowel quality reduction and are phonetically realised as [ɨ ɐ u], whereas the long vowels /$\overset{\mu\mu}{\text{ɛ}}$ $\overset{\mu\mu}{\text{a}}$ $\overset{\mu\mu}{\text{ɔ}}$/ — linked to two moras in the underlying representation — lose merely one of their moras, but retain their quality and are phonetically realised as [ɛ a ɔ]. Therefore, **the only information that speakers must store at the lexical level is theoretically whether or not a given lexical item contains an underlying long vowel**.

To summarise the above discussion: the phonological opposition in vowel quantity that once existed — as shown on the left-hand side of (8) — is still preserved in the underlying representation today, as an unpredictable and oppositive property of vowels; in contrast, the vowel alternations shown on the right-hand side of (8), both in quantity and in quality, are predictable from the presence or absence of stress.

A key clue in support of this hypothesis lies in the occasional occurrence of longer phonetic realisations of the vowels involved in the alternation [ɛ́ː áː ɔ́ː] ≈ [ɛ a ɔ], even in phonologically unstressed syllables. For instance, the root vowel in *p*[áː]*da* (< lat. ****pānāta***-) "small bread made with poor-quality flour" and its stress-shift counterpart *p*[a]*deiro* (<m.pt. ***paa****deiro* < Lat. ****pāna****tāriu*-) "(male) baker" and its derived form *p*[a]*deira* "(female) baker" is often realised as [aˑ] in *p*[aˑ]*deiro* and *p*[aˑ]*deira* — in contrast to the consistently short and close [ɐ] in *m*[ɐ]*deira* (< lat. ***mā****teriam*) "wood (material)"[2]. In our personal observation, the non-reduced vowel [a] occuring in unstressed syllables that function as the head of a metrical foot — as in *p*[a]*da*- of *p*[a]*daria* "bakery" ↼ *pada* + *-aria*, where ↼ indicates derivation — is often realised with a noticeably longer duration, as in *p*[aː]*daria*. This contrasts with cases like the anthroponym *C*[ɐ]*tarina* "a female given name, equivalent to English *Catherine*", where the head of the foot *C*[ɐ]*ta*- contains the reduced vowel [ɐ], which is clearly shorter.

The following are spectrographic analysis images of the audio files available for hearing on *Infopédia*, referenced in footnote 2. Figure (14) corresponds to *p**a**daria*, and Figure (15) to *C**a**tarina*. The analysis was conducted using Praat (version 64.3.5). Although no information is provided regarding the speaker, it is a native female speaker, and the same person for both recordings. The overall word durations are 739ms for *p*[aː]*daria* and 726ms for *C*[ɐ]*tarina*, and are thus nearly identical. Both words also share the same metrical structure — namely, (ˌ*p*[aː].*da*)(ˈ*ri.a*) and (ˌ*C*[ɐ].*ta*)(ˈ*ri.na*), where ( ) indicates a metrical foot, . a syllable boundary, ˈ and ˌ the primary and secondary stresses, respectively — and the two vowels under comparison both constitute the head of the secondary-stressed foot, not the primary-stressed position. In addition, the vowels that follow them are entirely identical: namely, [-ɐ-], [-íː-] and [-ɐ].

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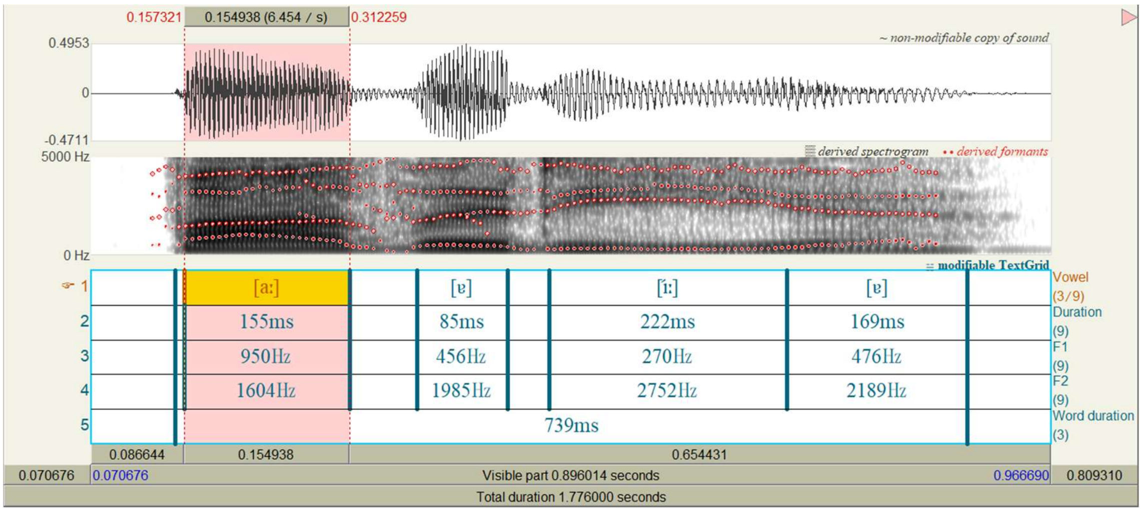
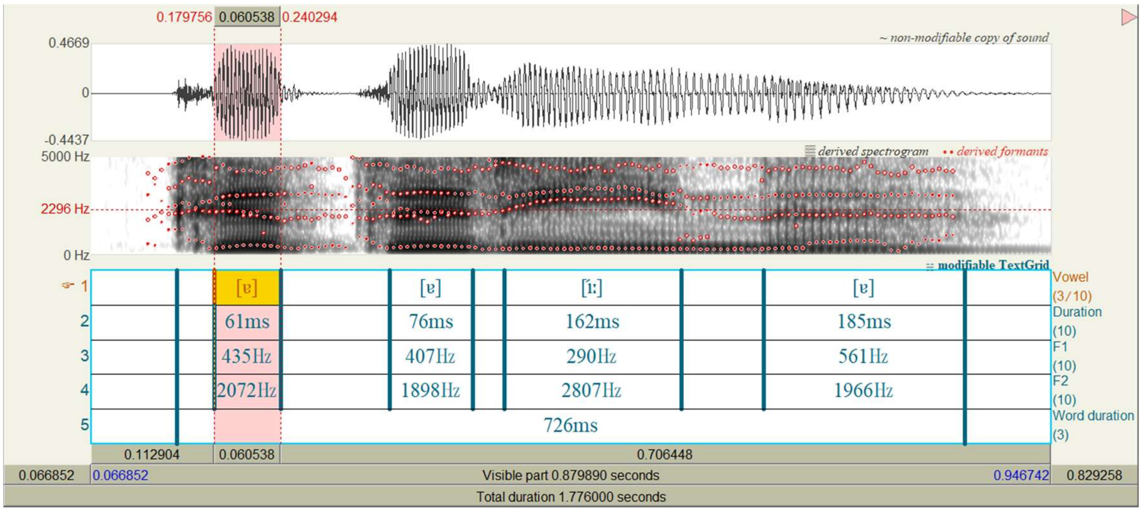
[2] As of 5 June 2025, the pronunciations of *pada*, *padeiro*, *padeira*, *madeira*, *padaria*, *Catarina, pegada*, and *pegado* can be accessed for hearing via the *Infopédia* online dictionary at the following entries:
https://www.infopedia.pt/dicionarios/lingua-portuguesa/pada
https://www.infopedia.pt/dicionarios/lingua-portuguesa/padeiro
https://www.infopedia.pt/dicionarios/lingua-portuguesa/padeira
https://www.infopedia.pt/dicionarios/lingua-portuguesa/madeira
https://www.infopedia.pt/dicionarios/lingua-portuguesa/padaria
https://www.infopedia.pt/dicionarios/lingua-portuguesa/catarina
https://www.infopedia.pt/dicionarios/lingua-portuguesa/pegada
https://www.infopedia.pt/dicionarios/lingua-portuguesa/pegado
The vowel [aː] in *p*[aː]*daria* is particularly long and warrants close auditory attention.

(14) ***pa**daria* [pa:ðɐˈɾi:ɐ]



(15) ***Ca**tarina* [**kɐ**tɐˈɾi:nɐ]



The acoustic measurements of the foot *pada-* in the noun *padaria* and of the foot *Cata-* in the noun *Catarina* are given in (16) and (17), respectively:

(16)

Syllable	Vowel	Duration (ms)	F1 (Hz)	F2 (Hz)
pa-	[a:]	155	950	1604
-da-	[ɐ]	85	456	1985

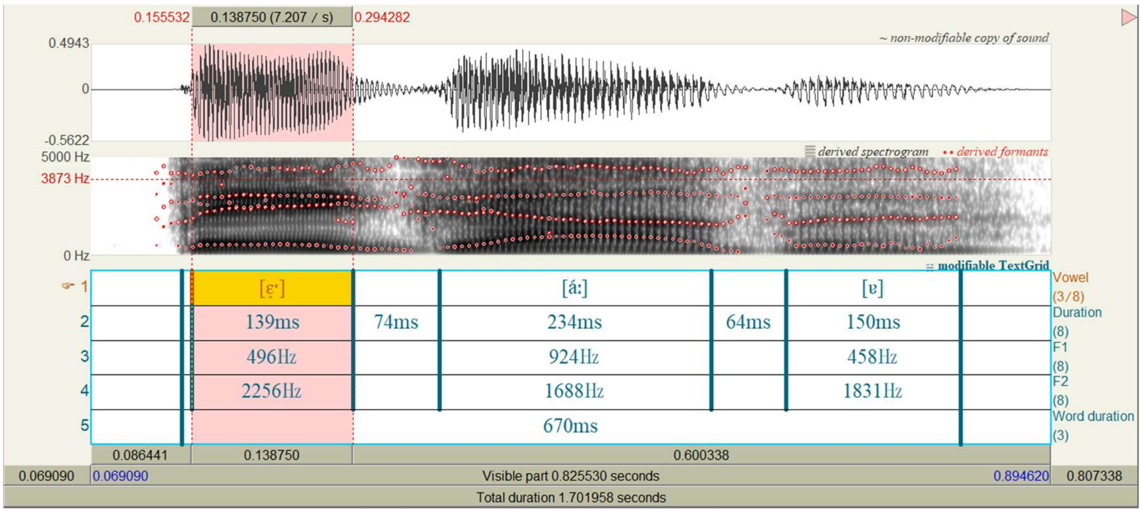
(17)

Syllable	Vowel	Duration (ms)	F1 (Hz)	F2 (Hz)
Ca-	[ɐ]	61	435	2072
-ta-	[ɐ]	76	407	1898

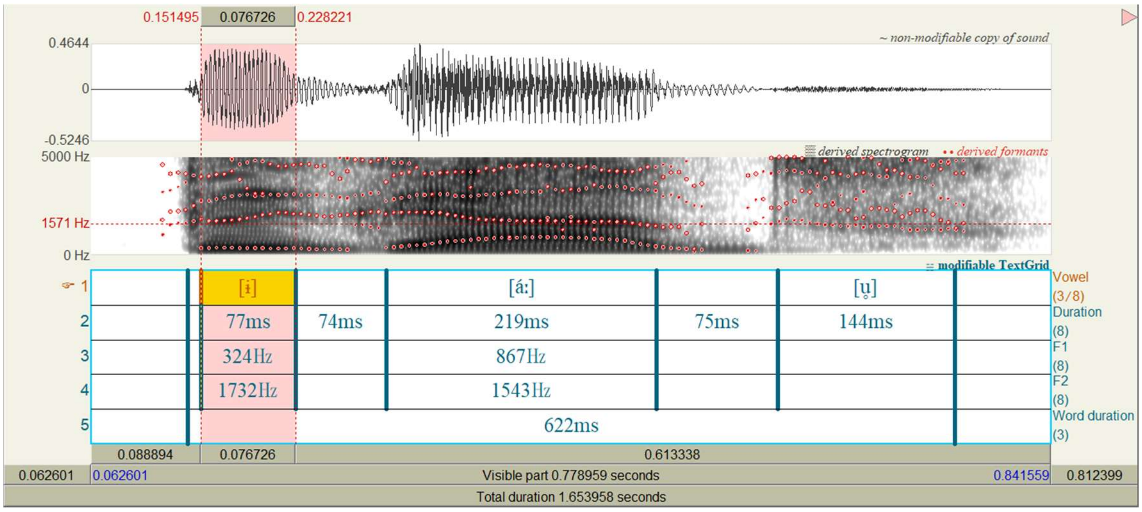
In (16), the vowel [aː] in *p*[aː]*daria* exhibits markedly longer duration (155ms) and a more open quality (F1 = 950 Hz) than the corresponding vowel [ɐ] in *C*[ɐ]*tarina* (61ms, F1 = 435 Hz) in (17). This striking contrast between *p*[aː]*daria* and *C*[ɐ]*tarina* is likely to provide strongly suggestive phonetic evidence that underlying bimoraic vowels /$\overset{\mu\mu}{\text{a}}$/ are preserved not only in quality but also in duration in non-stressed foot-head positions in Contemporary EP. This would suggest that the analysis is not merely hypothetical.

Next, (18) - (20) present acoustic data from the noun *pegada* (< m.pt. *peegada* < lat. ****pedicāta****-) "footprint" — derived from ***pé*** (< m.pt. ***pee*** < lat. ***pede***-) "foot" — which is assumed to have /$\overset{\mu\mu}{\varepsilon}$/, and *pegado*, from the verb *pegar* (< lat. *picāre*) "to hold", which is assumed to contain /$\overset{\mu}{\varepsilon}$/:

(18) *pegada* [p**ɛ**ˑˈgaːðɐ]



(19) *pegado* [p**ɨ**ˈgaːd̥ʷ]



(20)	Syllable	Vowel	Duration (ms)	F1 (Hz)	F2 (Hz)
	pe-	[ɛˑ]	139	496	2256

(21)	Syllable	Vowel	Duration (ms)	F1 (Hz)	F2 (Hz)
	pe-	[ɨ]	77	324	1732

The analysis was conducted in the same manner as (14) and (15). As far as we can tell, it is the same native female speaker. Both words share the same metrical structure — namely, (ˌ*p*[ɛˑ]) (ˈ*ga.da*) and (ˌ*p*[ɨ])(ˈ*ga.do*) — and the two vowels under comparison both constitute the head of the word-initial degenerate foot, not the primary-stressed position. Moreover, the vowels that follow them are almost identical: namely, [-áː-] and [-ɐ] or [-u]. In (18), the pretonic vowel [ɛˑ], assumed to be bimoraic here, in *p*[ɛˑ]*gada* exhibits nearly double the duration (148ms) and a mid front quality (F1/F2 = 496/2256 Hz) compared to the unstressed vowel [ɨ] in *p*[ɨ]*gado*, which is assumed to be monomoraic and shows a high central quality (85ms, F1/F2 = 324/1732 Hz) in (19). These behaviours in terms of vowel length suggest that the vowels showing the alternation [ɛ́ː áː ɔ́ː] ≈ [ɛ a ɔ] are underlyingly long vowels /$\overset{\mu\mu}{\text{ɛ}}$ $\overset{\mu\mu}{\text{a}}$ $\overset{\mu\mu}{\text{ɔ}}$/, which have a strong tendency to be phonetically realised as short vowels under lack of stress due to mora loss.

Another suggestive observation is that, in Contemporary EP, *length* serves as the phonetic implementation of stress (cf. Martins 1983). Accordingly, it is not surprising that **underlyingly short vowels /$\overset{\mu}{\text{V}}$/ tend to be phonetically lengthened when stressed**, while **underlyingly long vowels /$\overset{\mu\mu}{\text{V}}$/ tend to be phonetically shortened when unstressed**.

Finally, phonetic realisations of vowel coalescence at the phrasal level observed in Contemporary EP also may be taken as evidence supporting this interpretation. For example, the final vowel *-a* in *fic**a*** — the second person singular imperative form of the verb *ficar* "to stay" — and the initial vowel *a-* in the locative adverb ***a**qui* "here" are both unstressed, and when pronounced in isolation, each of these is typically realised as the reduced vowel [ɐ], as in *fic*[ɐ] and [ɐ]*qui*. When the two words are combined to form the imperative sentence *Fic**a** **a**qui* "Stay here", however, it is far more natural in ordinary speech to pronounce them not as *Fic*[**ɐ ɐ**]*qui*, but rather as *Fic*[**a**]*qui*, with the non reduced open vowel [a]. This synchronic vowel crasis can be accounted for by the diagram in (22), assuming the hypothesis proposed above — namely, that an opposition has persisted between the underlying short vowels /$\overset{\mu}{\text{ɛ}}$ $\overset{\mu}{\text{a}}$ $\overset{\mu}{\text{ɔ}}$ / and the underlying long vowels /$\overset{\mu\mu}{\text{ɛ}}$ $\overset{\mu\mu}{\text{a}}$ $\overset{\mu\mu}{\text{ɔ}}$/, which, in unstressed syllables, undergo the reduction /$\overset{\mu}{\text{ɛ}}$ $\overset{\mu}{\text{a}}$ $\overset{\mu}{\text{ɔ}}$/ → [ɨ ɐ u] and /$\overset{\mu\mu}{\text{ɛ}}$ $\overset{\mu\mu}{\text{a}}$ $\overset{\mu\mu}{\text{ɔ}}$/ → [ɛ a ɔ]:

(22) *Fic*$\overset{\mu}{\boldsymbol{a}}$ $\overset{\mu}{\boldsymbol{a}}$*qui* → *Fic*$\overset{\mu}{\cancel{\boldsymbol{a}}}$ $\overset{\mu}{\boldsymbol{a}}$*qui* → *Fic'*$\overset{\mu\mu}{\boldsymbol{a}}$*qui*

In more explicit terms, the final *a* of *fic*$\overset{\mu}{\boldsymbol{a}}$ and the initial *a* of $\overset{\mu}{\boldsymbol{a}}$*qui* are each underlying monomoraic vowels /$\overset{\mu}{\text{a}}$/. Therefore, when these words are pronounced in isolation, both vowels appear in unstressed position and undergo vowel quality reduction, being realised phonetically as [ɐ]. However, when the two words occur in sequence within a syntactic constituent, the adjacency of identical phonological elements is likely disfavoured by the OCP, leading to the deletion of the vowel quality of *a* in *fic*$\overset{\mu}{\boldsymbol{a}}$, while its associated mora is retained and relinked to the following $\overset{\mu}{\boldsymbol{a}}$- of $\overset{\mu}{\boldsymbol{a}}$*qui*. As a result, a *derived* long vowel $\overset{\mu\mu}{\text{a}}$ emerges in *fic'*$\overset{\mu\mu}{a}$*qui*, which is phonetically realised as a non-reduced open vowel [a] like an *underlying* long vowel /$\overset{\mu\mu}{\text{a}}$/. Other examples of the same type of coalescence — e.g. *dava-o* [ˈdaːvɔ] "he/she/you used to give it" and *dav**a-a*** [ˈdav**a**] "he/she/you used to give it" — will be dealt with in Section 6.3.

In sum, it is assumed that, in EP, the historical strengthening of stress — which is reflected in the fact that "O português vem sendo trabalhado desde séculos pela tendência ao enfraquecimento das vogais átonas" (Teyssier 1982: 66) — has led to a situation in which, **in stressed syllables, the lengthening of short vowels has neutralised the underlying opposition in vowel quantity, making it no longer perceptible at the phonetic level**. By contrast, **in unstressed syllables, the reduction of vowel quality in short vowels and the reduction of quantity in long vowels has caused the underlying opposition to be phonetically implemented by means of vowel quality**[3].

4. From underlying to surface representations

The next issue concerns how to account for the mapping of vowel segments from their underlying representations to their surface ones: /$\overset{\mu}{\text{ɛ}}$ $\overset{\mu}{\text{a}}$ $\overset{\mu}{\text{ɔ}}$/ → [ɛ́ː áː ɔ́ː] ≈ [ɨ ɐ u] and /$\overset{\mu\mu}{\text{ɛ}}$ $\overset{\mu\mu}{\text{a}}$ $\overset{\mu\mu}{\text{ɔ}}$/ → [ɛ́ː áː ɔ́ː] ≈ [ɛ a ɔ]. In addition to /$\overset{\mu}{\text{ɛ}}$ $\overset{\mu}{\text{a}}$ $\overset{\mu}{\text{ɔ}}$/, the underlying inventory of short vowels in Contemporary EP also includes /$\overset{\mu}{\text{i}}$ $\overset{\mu}{\text{e}}$ $\overset{\mu}{\text{o}}$ $\overset{\mu}{\text{u}}$/. Taking these into account, it appears that in stressed open syllables — particularly not before [+nasal] consonants, [−back, +high] consonants, nor vowels[4] — and in

[3] It may be observed that the present analysis, in adopting a moraic approach to the vowel system of EP, may in some respects parallel the corrective developments seen in Japanese phonology, where mora-based analyses — initially formulated by native speakers of Japanese, for whom, whereas the mora is self-evident, the syllable does not naturally and intuitively emerge as both a phonetic and phonological unit; indeed, for speakers of a language where the syllable is self-evident, it must be surprising that it is practically impossible for the average native speaker of Japanese to identify or count syllables in speech without explicit instruction — have gradually been balanced and refined by syllabic perspectives through external reinterpretations. The present proposal likewise seeks not to import foreign models, but rather to uncover the underlying prosodic logic implicit in the EP system itself.

[4] In Portuguese, the syllable-initial [+nasal] consonants are [m n ɲ]. In this paper, syllable-initial [ʃ ʒ ʎ ɲ] are considered doubly articulated consonants, specified as [+high, −back] in their secondary articulation, as shown in Section 6.2.

pretonic open syllables — especially when not in absolute word-initial position — the following complex mapping generally occurs: $/\overset{\mu}{\text{i}}\ \overset{\mu}{\text{e}}\ \overset{\mu}{\text{ɛ}}\ \overset{\mu}{\text{a}}\ \overset{\mu}{\text{ɔ}}\ \overset{\mu}{\text{o}}\ \overset{\mu}{\text{u}}/$ → [íː éː ɛ́ː áː ɔ́ː óː úː] ≈ [i ɨ ɨ ɐ u u u]. This paper proposes that these mappings, including those described earlier, can be systematically derived within the framework of Stratal Optimality Theory (Stratal OT), as shown in (23):

(23)

μ μ μ μ μ μ μ /i e ɛ a ɔ o u/	μ μ μ μ μ μ μ /i e ɛ a ɔ o u/	μμ μμ μμ /ɛ a ɔ/	μμ μμ μμ /ɛ a ɔ/	Underlying representation
í é ɛ́ á ɔ́ ó ú	—	**ɛ́ː áː ɔ́ː**	—	**Stress assignment**
—	**i ɨ ɨ ɐ u u u**	—	—	**Vowel quality modification**
íː éː ɛ́ː áː ɔ́ː óː úː	—	—	**ɛ a ɔ**	**Vowel quantity adjustment**
[íː éː ɛ́ː áː ɔ́ː óː úː]	[i ɨ ɨ ɐ u u u]	[ɛ́ː áː ɔ́ː]	[ɛ a ɔ]	Surface representation

Since the reduction of the vowel quality of underlying short vowels occurs in unstressed syllables, syllabification and stress assignment must already have taken place prior to that reduction stage. Furthermore, since vowel quality reduction in unstressed syllables affects only underlying short vowels, the underlying distinction in vowel quantity must still be preserved at that stage. Vowel quantity adjustment, then, must take place only subsequently. Under the current model, therefore, the following strata and constraint rankings are being hypothesised:

(24) Lexical Stratum 1: Feature Establishment

At this stratum, after word formation, the feature values of each segment are re-specified. It is also at this stage that floating mora linking is processed, as in $\mathit{f\overset{\mu}{a}l\text{-}\overset{\mu}{\mathbf{a}}\text{-}\overset{\mu}{\ }\,m\overset{\mu}{o}s} \rightarrow \mathit{f\overset{\mu}{a}l\text{-}\overset{\mu\mu}{\mathbf{a}}\text{-}m\overset{\mu}{o}s}$ — the first person plural form *falámos* "we spoke" in the preterite indicative of the verb *falar* "to speak" — and in $\mathit{in\text{-}l\overset{\mu}{\mathbf{a}}t\text{-}\overset{\mu}{0}\text{-}iv\text{-}o} \rightarrow \mathit{i\text{-}l\overset{\mu\mu}{\mathbf{a}}t\text{-}0\text{-}iv\text{-}o}$ —the adjective *ilativo* "inferential", as shown in Sections 6.4 and 6.5.

(25) Lexical Stratum 2: Syllabification

(26) Lexical Stratum 3: Stress Assignment

As Makino (2010: ; 2012:) stated: "In the default stress-assignment process, the final syllable of a word — which contains the vowel *-o*, *-a*, *-e*, or the vowel *-0* with no segmental quality but with a vocalic quantity of one mora — serves as the starting point for binary foot construction proceeding from right to left. In this process, each foot is trochaic, consisting of a stressed syllable followed by an unstressed one, and the stressed syllable of the rightmost foot is assigned as the word's main stress syllable". By contrast, "the marked stress-assignment process is triggered by lexical-exceptional morphemes which, when adjacent to the final syllable of the word, require their own final syllable to serve as the starting point for binary foot construction. In such cases, footing proceeds irregularly from the penultimate syllable, and the final syllable is rendered extrametrical and thus remains unstressed" (both passages translated from Japanese).

(27) Postlexical Stratum 1: Vocalic Quality Modification

In this stratum, the following set of constraints is assumed to be interacting:

The inviolable constraint IDENT-QUANTITY;

IDENT(low∨high) ≫ *[+low]/V[−stressed, μ] ≫ *[−high]/V[−stressed, μ] ≫ IDENT(high);

*[+low]/V[−stressed, μ] ≫ IDENT(low);

*[±*ATR*]/V[−stressed, μ] ≫ IDENT(*ATR*);

*[−back]/V[−stressed, μ]__([−sonorant])# ≫ IDENT(high∧back) ≫ *[−back]/V[−stressed, μ] ≫ IDENT(back);

IDENT(high), IDENT(low), IDENT(*ATR*), IDENT(round) ≫ *[+high], *[−high], *[+low], etc.

(28) Postlexical Stratum 2: Vocalic Quantity Adjustment

In this stratum, the following set of constraints is assumed to be interacting:

The inviolable constraint IDENT-QUALITY;

MAX, DEP ≫ *μμμ ≫ *ˈσ[μ] ≫ *μμ/V ≫ MAX(μ);

*ˈσ[μ] ≫ *μμ/V, DEP(μ), DEP(high), DEP(*ATR*)

*LONG-MID-FRONT_.V ≫ DEP(high), DEP(*ATR*)

*NON-HIGH-OFFGLIDE ≫ DEP(high), DEP(*ATR*)

While Lexical Stratum 1 (Feature Establishment), Lexical Strata 2 (Syllabification), and 3 (Stress Assignment) are important background, they fall beyond the primary focus of this article. We shall therefore defer a full formalization of those levels to a subsequent paper. The present study shall concentrate exclusively on:

Postlexical Stratum 1: Vocalic Quality Modification;

Postlexical Stratum 2: Vocalic Quantity Ajustment.

4.1. Postlexical Stratum 1: vocalic quality modification

In this stratum, the short vowels $\overset{\mu}{\text{i}}\ \overset{\mu}{\text{e}}\ \overset{\mu}{\text{ɛ}}\ \overset{\mu}{\text{a}}\ \overset{\mu}{\text{ɔ}}\ \overset{\mu}{\text{o}}\ \overset{\mu}{\text{u}}$ — hereafter simplified as i, e, ɛ, a, ɔ, o, u — generally undergo changes in vowel quality in *unstressed* syllables. The present study limits its discussion to the following types of unstressed syllables: non-absolute word-initial, word-medial and word-final open syllables, as well as non-absolute word-initial, word-medial and word-final syllables closed by /s z/, and non-absolute word-initial and word-medial syllables closed by /r/[5]. By contrast, the long vowels ɛː aː ɔː pass through this stratum without undergoing any changes in either quality or quantity, and will instead be subject to quantitative adjustment in the next stratum, which will be discussed in Section 4.2. Before turning to the feature value changes affecting the short vowels in this stratum, however, it is necessary to first consider the nature of tongue-height or aperture oppositions among the vocalic segments of Contemporary EP.

In representative generative phonological studies of Portuguese — namely, Andrade (1977: 96-100), Mateus (1982: 25-26), Mateus & Andrade (2002: 31–35), and Mateus et al. (2003: 998–1001) — an analysis such as (29) has been adopted[6]. In this framework, the feature [−low] is considered redundant for [+high] vowels, just as [−high] is considered redundant for [+low] vowels.

(29)

high				low
+	/i/	[ɨ]	/u/	−
−	/e/	[ɐ]	/o/	
	/ɛ/	/a/	/ɔ/	+

This analysis of vowel tongue-height or aperture is simple and free of redundancy, but it

[5] In unstressed syllables closed by /l/, the vowel reduction described in (23) does not occur (e.g. *feltar* "to felt" /feltr-a-r0/ → [feɫˈtɾaːɾ]; *selvagem* "wild" /sɛlv-aʒen/ → [sɛɫˈvaːʒɐ̃i̯]; *faltar* "to lack" /falt-a-r0/ → [fɑɫˈtaːɾ]; *voltar* "to return" /vɔlt-a-r0/ → [voɫˈtaːɾ]~[vɔɫˈtaːɾ]; *amável* "kind" /am-a-vɛl-0/ → [ɐˈmaːvɛɫ]; *Setúbal* "a Portuguese city's name" /setubal-0/ → [sɨˈtuːβɑɫ]). In unstressed syllables closed by /n/, the vowel+/n/ sequence — which bears two moras — is assumed to undergo vocalic nasalisation and nasal deletion in an earlier stratum — e.g. /in/ → ĩː — and thus enters the current stratum as a bimoraic nasal vowel, as discussed in Section 6.1. Other environments — namely, absolute word-initial unstressed open syllables as well as absolute word-initial unstressed syllables closed by /s/, /r/, or /n/ and word-final syllables closed by /r/ or /n/ — exhibit weakening patterns that vary depending on whether the syllable is open or closed, and on the nature of the coda consonant; due to the considerable complexity involved, these cases will not be discussed here.

[6] In Mateus & Andrade (2002), /i/ is treated as the least marked vowel in Portuguese and is subject to underspecification. As a result, /i/ and /u/ are not specified for either [high] or [low] in the underlying representation, and /e/ and /o/ are specified only as [−high], while /ɛ/, /a/, and /ɔ/ are specified only as [+low].

presents several problems in terms of phonological theory[7]. First, if /ɛ a ɔ/ formed a natural class on the basis of sharing the feature [−high, +low] in terms of vowel tongue-height or the correlate aperture, then from the perspective of phonological theory, one would expect them to behave uniformly with respect to processes of vowel raising in unstressed syllables, since phonological features are theoretical constructs introduced precisely to account for the common behaviour of a set of segments. In pretonic CV syllables, /a/ is raised to the mid vowel [ɐ], specified as [−high, −low]. If /ɛ/ and /ɔ/ formed the same natural class with /a/, one would expect them to be raised — with the feature value change [+low] → [−low] — to the mid vowels [e] and [o], likewise specified as [−high, −low], as is [ɐ]. Nevertheless, they are actually raised to the high vowels [ɨ] and [u] — that is, vowels specified as [+high, −low], respectively. This difference in pattern suggests that these three vowels do not belong to the same natural class in terms of tongue-height or aperture.

Second, in EP, /a/ undergoes raising to [ɐ́ː] in stressed CV syllables preceding [m n ɲ], whereas /ɛ/ and /ɔ/ retain their underlying tongue-height or aperture and are realised as [ɛ́ː] and [ɔ́ː] — e.g. *cama* "bed" /kam-a/ → [ˈkɐːmɐ]; *remo* "oar" /rɛm-o/ → [ˈʀɛːmu]; *fome* "hunger" /fɔm-e/ → [ˈfɔːmɨ]. Within the aforementioned model, this leads to a further asymmetry: although /ɛ a ɔ/ are all analysed as [+low] and thus form a natural class, /a/ behaves differently from /ɛ/ and /ɔ/ with respect to another type of raising. Consequently, if one assumes that /ɛ a ɔ/ form a natural class by virtue of their [+low] specification, one must then explain why /a/ raises while /ɛ/ and /ɔ/ do not in this context, based on some additional feature that distinguishes /a/ from the other two within the [+low] set. In their descriptive framework, /ɛ/ is distinguished from /a/ by its [−back] specification, and /ɔ/ is distinguished from /a/ by its [+round] specification. It would therefore follow that the features [−back] in /ɛ/ and [+round] in /ɔ/ somehow inhibit the [+low] → [−low] change from applying in stressed CV syllables preceding [+nasal] consonants. But this seems to be an implausible explanation.

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[7] The reason why an analysis such as that shown in (29) has been adopted likely lies in the fact that, when following Halle's articulator-based model and analysing [i u] as [+high], the set [e o ɛ ɔ] as [−high, −low], and [a] as [+low], there is no alternative but to define the opposition between [e o] and [ɛ ɔ] in terms of [+ATR] versus [−ATR] opposition. For native researchers, however, the opposition between [e o] and [ɛ ɔ] — as suggested by orthographic conventions, that is, the acute and circumflex accents (á/â, é/ê, ó/ô), the denomination of "vogais fechadas/semifechadas/semiabertas/abertas", and their own phonological analyses — is intuitively one of tongue-height or aperture, and it is likely that there has been hesitation to introduce [±ATR], in its original sense of tongue root advancement, into the analysis of their own native language.[8] The vowels /a/ and /i u/ stand in a binary opposition in terms of tongue-height or aperture. Accordingly, if this opposition is to be represented by features that may assume both positive and negative values, a single feature is sufficient, and employing two features would be theoretically redundant. It should be noted that, in this model, the starting point is not taken to be [±high], because — as will become clear in the next paragraph — doing so would require describing /e o/ as [+high, +low], thereby breaking terminological continuity with the traditional model.

Third, in pretonic CV syllables in Brazilian Portuguese — henceforth, BP — /a/ remains as [a], specified as [+low], while /ɛ/ and /ɔ/ are raised to the mid vowels [e] and [o], specified as [−low], respectively — at least from the perspective of the aforementioned framework. This too is difficult to account for under the analysis that treats /ɛ a ɔ/ as forming a natural class of [+low] vowels. On the other hand, if /a/ is analysed as [+low] and /ɛ ɔ/ as [−low, −high, −*ATR*] — where [*ATR*] in italics, as we shall see later in this section, does not refer to literal tongue root advancement — then the three vowels do not form a natural class, and there is no reason to expect them to pattern together phonologically with respect to tongue-height or vocal tract aperture. Moreover, this raising can be analysed as involving a shift in feature specification: [−*ATR*] → [+*ATR*] in the case of /ɛ ɔ/ → [e o].

Fourth, in BP, the word-final sequences /-e-a/ and /-ɛ-a/ are phonemically and phonetically opposed, as exemplified by *feia* "ugly" /fe-a/ → [ˈfei̯.ɐ] and *ideia* "idea" /idɛ-a/ → [iˈdɛi̯.ɐ], respectively. In EP, however, this opposition between stressed vowels is phonetically neutralised: both /-e-a/ and /-ɛ-a/ are realised as [-ɐ́i̯.ɐ], as in *feia* /fe-a/ → [ˈfɐi̯.ɐ] and *ideia* /idɛ-a/ → [iˈðɐi̯.ɐ]. That is, [ɐ] functions as a central **mid** vowel that corresponds to both the front **close-mid** vowel /e/ and the front **open-mid** vowel /ɛ/ and, as such, it stands in opposition to the central **open** vowel /a/, as in *má* "bad" /ma-a/ → [ˈmaː] and *faia* "beech" /fai-a/ → [ˈfai̯.ɐ]. The centralisation [−back] → [+back] of /e/ and /ɛ/ before seemingly epenthetic [i̯] may be attributed to a restriction on the sequence of [−back] features, enforced by the OCP, since /e/, /ɛ/, and [i̯] are all specified as [−back]. The fact that both vowels are realised as the same allophone [ɐ] and thus neutralised suggests that, within the [+back, −round] region, the [±*ATR*] opposition — as a feature encoding *tongue-height* or *aperture* — is not available as a distinctive specification in the system. If /ɛ/ were [+low] just like /a/, there would be no explanation for why *ideia* is not realised as *[iˈðai̯.ɐ] — namely, /idɛ-a/ → iˈðɛi̯.ɐ → *[iˈðai̯.ɐ], simply with the change [−back] → [+back].

From a theoretical perspective, it is reasonable to posit that the low vowel /a/ occupies the foundational position in the vowel system — articulatorily the most low and acoustically the most salient vowel. The high vowels /i/ and /u/ then arise as maximally opposed poles relative to /a/, completing a triangular system that ensures maximal perceptual opposition. Subsequently, the addition of /e/ and /o/ serves to fill the **mid**-space between these extremes, producing the commonly observed five-vowel system. Only later, as a form of system refinement, do languages introduce further oppositions between **close-mid** and **open-mid** vowels — /e/ vs. /ɛ/, and /o/ vs. /ɔ/ — distinctions that are structurally fragile and often subject to neutralisation as in

EP. This trajectory — from /a/ to /i u a/, then to /i u e o a/, and finally to /i u e o ɛ ɔ a/ — appears to align closely with both the gradual subdivision of the articulatory-acoustic vowel space and with cross-linguistic patterns of oppositional vowel relationships.

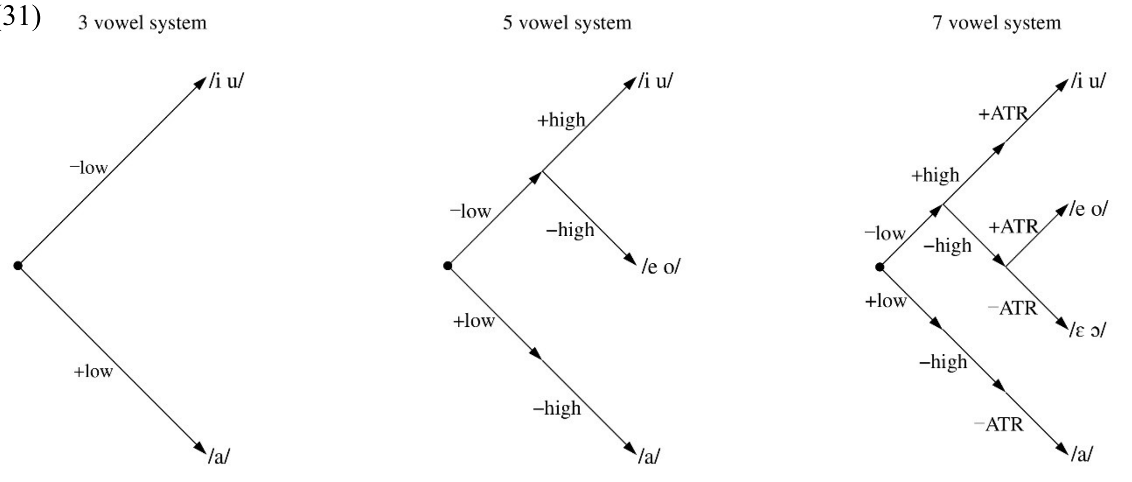
Based on the above discussion, then, it seems more appropriate to analyse the vowel tongue-height opposition among EP vocalic segments in terms of [±high] and [±low], as follows:

(30)

high				low
+	/i/	[ɨ]	/u/	−
−	/e/	[ɐ]	/o/	
	/ɛ/		/ɔ/	
		/a/		+

The problem with this system (30) is that it lacks a feature capable of representing the opposition — in tongue-height or aperture — between /e o/ and /ɛ ɔ/, as indicated by the shaded cells. Given that the aforementioned studies consistently analyse this opposition as one between [−low] and [+low], and make no reference to [±ATR] — and moreover, other researchers within the structuralist–functionalist tradition have likewise consistently treated this as an opposition of aperture — it appears that, for native-speaker researchers, this is self-evidently a tongue-height-based or aperture-based opposition. Accordingly, in this paper, the feature [±*ATR*] — which is generally assumed to be dominated by the Pharyngeal Cavity node in the feature geometry — is employed in italics as a convenient label for representing **an additional tongue-height or aperture opposition within the [−high, −low] region**. This paper proposes the following (31) systems of vowel tongue-height:

(31)



In (31), [±low], [±high], and [±ATR] are reinterpreted as features that encode the conceptual movement of the tongue for vowel segments within the vowel space — starting from a neutral position, represented by ● — toward their specific tongue-height targets within the system: high /i, u/, mid-high /e, o/, mid-low /ɛ ɔ/, and low /a/.

First, [−low] and [+low] are taken to encode upward and downward movement, respectively, from the neutral position. Whereas in traditional systems [−low] expresses a neural command not to lower the tongue from the neutral position and, correlatively, the tongue position itself, in the present system it represents upward movement from that position. In a three-vowel system such as /i u a/, the endpoints of the [±low] movement coincide with the upper and lower bounds of the vowel space, thereby determining the tongue-height — or aperture — of each vowel: /i u/ are specified as [−low], while /a/ is specified as [+low][8].

By contrast, in a five-vowel system such as /i u e o a/, the feature [±low] alone does not suffice to determine the tongue-height of each vowel. From the endpoint of [±low] movement, a subsequent [±high] movement begins: [+high] indicates upward movement, and [−high] indicates downward movement. Thus, /i u/ reach their respective endpoints by undergoing a continuous upward movement specified first by [−low] and then by [+high], while /a/ reaches its endpoint through a continuous downward movement specified by [+low] followed by [−high]. In both cases, the endpoint of the movement coincides with either the upper or lower boundary of the vowel space, thereby terminating the motion. In contrast, the mid vowels /e o/, specified as [−low, −high], first rise under [−low], then change direction under [−high], resulting in an endpoint located at an intermediate tongue-height between that of /i u/ and /a/. Note, however, that in actual articulation, the tongue does not first rise and then fall for /e o/; rather, they are produced directly at a mid-tongue position. The sequence [−low, −high] represents not a literal articulatory trajectory but rather the abstract vectorial structure by which each vowel segment's tongue-height is determined. The markedness of mid vowels /e o/ in relation to both the high vowels /i u/ and the low vowel /a/ is thereby captured by the fact that the vector leading to their endpoint includes a point of inflexion — that is, a structural complexity.

Furthermore, in a seven-vowel system such as /i u e o ɛ ɔ a/, the features [±low] and [±high] alone are insufficient to determine the tongue-height of each vowel. From the endpoint of the [±high] movement, a further movement specified by [±*ATR*] begins. [+*ATR*] represents an

[8] The vowels /a/ and /i u/ stand in a binary opposition in terms of tongue-height or aperture. Accordingly, if this opposition is to be represented by features that may assume both positive and negative values, a single feature is sufficient, and employing two features would be theoretically redundant. It should be noted that, in this model, the starting point is not taken to be [±high], because — as will become clear in the next paragraph — doing so would require describing /e o/ as [+high, +low], thereby breaking terminological continuity with the traditional model.

upward movement, while [−*ATR*] represents a downward movement. As already noted, it is important to bear in mind that in this paper, [*ATR*] in italics does not refer to tongue root advancement in the narrow sense. Accordingly, /i u/ undergo a continuous upward movement in the sequence [−low, +high, +*ATR*], reaching their final endpoint; /a/, on the other hand, undergoes a continuous downward movement in the sequence [+low, −high, −*ATR*], likewise reaching its endpoint. In both cases, the endpoint coincides with the upper and lower bounds of the vowel space, and thus the movement terminates. The vowel /e o/, specified as [−low, −high, +*ATR*], initially ascends by [−low], then changes direction to descend via [−high], and then ascends again via [+*ATR*], ultimately reaching a mid-close tongue-height as the endpoint. By contrast, /ɛ ɔ/, specified as [−low, −high, −*ATR*], first ascend via [−low], then descend via [−high], and then continue to descend via [−*ATR*], ultimately reaching a mid-open tongue-height as the endpoint. In summary:

/a/ [+low, −high, −*ATR*]: **neutral position → descent → descent → descent**

/i u/ [−low, +high, +*ATR*]: **neutral position → ascent → ascent → ascent**

/ɛ ɔ/ ([−low, −high, −*ATR*]): **neutral position → ascent → descent → descent**

/e o/ ([−low, −high, +*ATR*]): **neutral position → ascent → descent → ascent**

To summarise the above, each of the features [±low], [±high], and [±ATR] corresponds to a single segment of the movement through which a vowel's tongue-height — and consequently its aperture — is determined. Rather than functioning as coordinates that statically locate the vowel within the vowel space, these features behave as vectors that define the directional trajectory toward the vowel's articulatory endpoint.

What must be noted here is the asymmetry in markedness between the mid vowels /e o/ and /ɛ ɔ/ in the context of a seven-vowel system. While both sets of vowels are marked in that their tongue-height is determined by a vector with an inflection point, /ɛ ɔ/ are less marked than /e o/ in that they involve only one inflection point, whereas /e o/ involve two. From this perspective, the vowels in a five-vowel system that correspond to /e o/ are not /e o/ themselves, but rather /ɛ ɔ/. In Contemporary EP, when scholarly words of Latin origin contain stressed vowels written as *e* or *o*, their quality is consistently [ɛ] and [ɔ] and not [e] and [o], regardless of vowel length in the Latin source. In other words, the opposition between /e o/ and /ɛ ɔ/ is neutralised in scholarly words. The fact that the outputs in such cases are [ɛ ɔ] — [−low, −high, −ATR] — and not [e o] — [−low, −high, +*ATR*] — may be attributed to the relative unmarkedness of [ɛ ɔ], whose defining vectors contain only a single inflection point.

Barbosa (1965: 76) notes that "/ɑ/ est probablement légèrement plus fermé dans *sei* «je sais» que dans *cama* «lit»", and this, too, can be accounted for within the framework of the schema in (31). The vowel /ɑ/ — namely [ɐ], as he considers the stressed *-e-* in *sei* and the stressed *-a-* in *cama* to be phonetic realisations of the same phoneme /ɑ/ — can be understood, in the case of *s*[ɐ́]*i*, as an allophone that results from the following process:

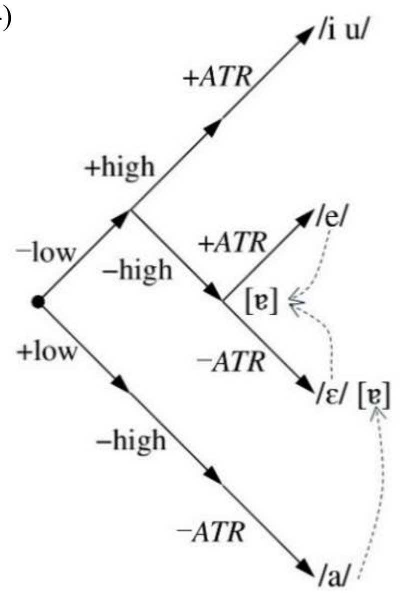
(32) The feature [−back], shared by /e/ specified as [−low, −high, +*ATR*, −back, −round] and /ɛ/ as [−low, −high, −*ATR*, −back, −round] is prohibited by the OCP from occurring adjacent to the same [−back] feature of a following [-i̯]. As a result, /e/ and /ɛ/ undergo a change from [−back] to [+back].

(33) Through the shift [−back] → [+back], /e/ and /ɛ/ move into the central mid vowel region, defined by the feature set [−low, −high, +back, −round]. In this region, the distinctive opposition in [±*ATR*] becomes impossible to maintain, leading to the deletion of the [±*ATR*] specification from both segments. As a result, they are both realised as [ɐ], a vowel lacking any *ATR* specification and characterised by **[−low, −high, +back, −round]**.

The forms discussed above — *feia* /f**e**-a/ → [ˈf**ɐ**i̯.ɐ] and *ideia* /id**ɛ**-a/ → [iˈð**ɐ**i̯.ɐ] — are precisely examples of this neutralisation of opposition.

On the other hand, the [ɐ] in *c*[ɐ́ː]*ma*, as examined in Section 6.4, is an allophone resulting from nasal resonance in the /a/, triggered by the spreading of the [+nasal] feature of [m], which diverts part of the pulmonic airstream into the nasal cavity. This reduces the amount of airflow available for oral articulation and renders the [+low] region unusable. Consequently, in the underlying form /a/ — specified as [+low, −low, −*ATR*, +back, −round] — only the feature [+low] is altered, changing to [−low], while all other features remain unaffected. Thus, the resulting [ɐ] in this context is specified as **[−low, −high, −*ATR*, +back, −round]**.

(34)



That this [ɐ] and the one derived from the neutralisation of the opposition between /e/ and /ɛ/ differ in tongue height is evident from (34). Furthermore, as we shall see immediately below, the interpretation of tongue-height or aperture degree in (31) also allows for a straightforward explanation of the tongue-height involved in the reduction of /a/ to [ɐ] in unstressed syllables.

Returning to the feature changes affecting the short vowels in this stratum, they are illustrated in (35) below. The altered feature values are shaded for clarity:

(35)

	high	low	*ATR*	back	round			high	low	*ATR*	back	round
i	+	−	+	−	−		i	+	−	0	−	−
e	−	−	+	−	−		ɨ	+	−	0	+	−
ɛ	−	−	−	−	−		ɨ	+	−	0	+	−
a	−	+	−	+	−	→	ɐ	−	−	0	+	−
ɔ	−	−	−	+	+		u	+	−	0	+	+
o	−	−	+	+	+		u	+	−	0	+	+
u	+	−	+	+	+		u	+	−	0	+	+

The underlying four degrees of tongue-height — /i u/ vs. /e o/ vs. /ɛ ɔ/ vs. /a/ — are reduced to a two-way opposition in unstressed position: [i ɨ u] vs. [ɐ]. Since [+low] is no longer active in the system, and all vowels involved share [−low], the resulting opposition in tongue-height is effectively reduced to a two-way contrast expressed solely by [±high]. Thus, the specification [±high] alone suffices, and [±*ATR*] becomes unnecessary and is left unspecified. In fact, the unstressed vowel [ɐ] differs in aperture from the stressed vowel [ɐ́], the latter of which is characterised as [−*ATR*]. Specifically, unstressed [ɐ] has an aperture closer to that of IPA [ə], a more closed central vowel. Emiliano (2009: 31–33) provides crucial observations on this point: while stressed [ɐ́] aligns in height with [ɛ] and [ɔ], unstressed [ɐ] is slightly higher and should be transcribed as [ə], positioned between [e o] and [ɛ ɔ] in terms of aperture. For example, the word *cama* should ideally be transcribed not as [ˈkɐm**ɐ**] but as [ˈkɐm**ə**].

In this weak position, vowel reduction is understood to proceed as follows:

(36) The centralisation of the low vowel /a/ occurs, resulting in /a/ → [ɐ], which corresponds to the feature change [+low] → [−low].

(37) As a consequence of (36), /e ɛ/, which share [−low, −high, −round] with the resulting [ɐ], undergo a feature change from [−high] to [+high] in order to preserve their underlying opposition with /a/, now realised as [ɐ].

(38) Furthermore, in order to maintain their underlying opposition to /i/, which surfaces as [i], /e ɛ/ also shift from [−back] to [+back], ultimately mapping /e ɛ/ to [ɨ].

(39) The feature [±*ATR*], which underlyingly distinguishes /e/ from /ɛ/, is based in EP on a

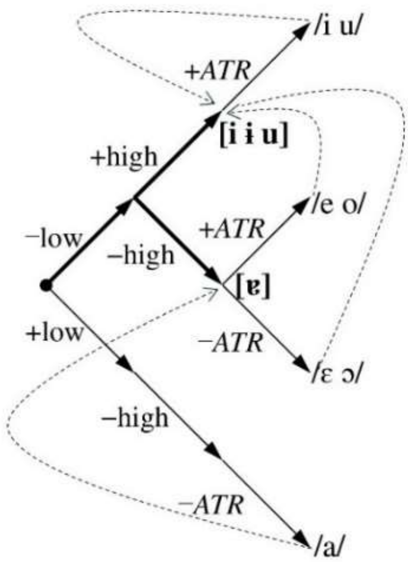
fine-grained tongue-height opposition among segments sharing [−low, −high]. Once /e ɛ/ become [+high], this opposition becomes untenable, and [±*ATR*] is accordingly neutralised: [±*ATR*] → [0*ATR*].

(40) The round back vowels /o ɔ/, which are specified as [−low, −high] and thus form a natural class with /e ɛ/ in terms of tongue-height, undergo parallel changes: [−high] → [+high] and [±*ATR*] → [0*ATR*], and are therefore mapped to [u].

(41) In order for /o ɔ/ now realised as [u] to preserve its opposition to the underlying /u/ realised also as [u], it would need furthermore to change [+round] to [−round]; however, doing so would yield [ɨ], thereby neutralising the underlying opposition to /e ɛ/ also realised as [ɨ]. As a result, the system instead opts to neutralise the opposition between /o ɔ/ and /u/, maintaining [u] in both cases.

This vowel reduction can be schematised as in (42), where the dashed curve represents the process of vowel reduction:

(42)



As we shall see immediately below, the conception of tongue-height degree or aperture degree introduced in (31) also enables a straightforward explanation of vowel reduction in unstressed syllables. In terms of tongue-height, this reduction is extremely simple: it consists merely in deleting all [*ATR*] values and raising the height by one step — [+low] becomes [−low] for /a/, and [−high] becomes [+high] for /e o ɛ ɔ/. For /e ɛ/, this change is accompanied by a shift from [−back] to [+back]. Accordingly, the vowel quality changes listed in (35) can be classified into two distinct types. The first type concerns features related to vowel height:

(43) Raising of mid vowels: [−high] → [+high] (e ɛ o ɔ → ɨ ɨ u u)

(44) Raising of low vowels: [+low] → [−low] (a → ɐ)

(45) Non-utilisation of [*ATR*] within the system: [±*ATR*] → [0 *ATR*] (i e ɛ a ɔ u → i ɨ ɨ ɐ u u u)

The second type pertains to the feature [back]:

(46) Centralisation of front mid vowels: [−high, −low, −back] → [+back] (e ɛ → ɨ)

From (43) and (44), the descriptive generalisation (47) can be formulated:

(47) In unstressed syllables, non-high short vowels specified as [−high] and low short vowels specified as [+low] are prohibited. This requirement is enforced by the unfaithful mappings [−high] → [+high] and [+low] → [−low] (e.g. *bolacha* "biscuit" /b**o**l-aʃ-a/ → b**o**ˈla.ʃa → b**u**ˈla.ʃɐ vs. *b**o**ˈla.ʃɐ; *bolacha* boˈla.ʃ**a** → buˈla.ʃ**ɐ** vs. * buˈla.ʃ**a**). However, the mappings [−high] → [+high] and [+low] → [−low] do not occur simultaneously; rather, only the latter takes place.

From this, the following constraint rankings can be inferred:

*[−high]/V[−stressed, μ] ≫ IDENT(high)

*[+low]/V[−stressed, μ] ≫ IDENT(low)

*[+low]/V[−stressed, μ] ≫ *[−high]/V[−stressed, μ]

The definitions of the above constraints are as follows:

***[−high]/V[−stressed, μ]**: Assign one * for each unstressed non-high short vowel.

***[+low]/V[−stressed, μ]**: Assign one * for each unstressed low short vowel.

IDENT(high): Assign one * for each change in the [high] feature between input and output.

IDENT(low): Assign one * for each change in the [low] feature between input and output.

This ranking alone cannot prevent, however, a candidate that simultaneously undergoes the mappings [−high] → [+high] and [+low] → [−low] — such as *bolacha* boˈla.ʃ**a** → *buˈla.ʃ**ɨ** — from incorrectly winning over the expected output buˈla.ʃ**ɐ**. To resolve this, a higher-ranked faithfulness constraint must be introduced above the ranking *[+low]/V[−stressed, μ] ≫ *[−high]/V[−stressed, μ]. This constraint should penalise the simultaneous violations of both IDENT(low) and IDENT(high) — as in the mapping a → *ɨ — while still favouring the mapping a → ɐ, which only violates IDENT(low). In doing so, the optimal candidate ɐ will correctly emerge as the winner over *ɨ.

That is, the required constraint is one that **allows for a change in either the [+low] or [−high] feature independently, but prohibits simultaneous changes in both**. This constraint may be formalised as:

IDENT(low⋁high): Assign one violation mark if both [low] and [high] features change their values simultaneously.

This constraint is **indispensable for ensuring that /a/ preserves its underlying opposition with /e ɛ/, which surface as [ɨ]**. From this, the following constraint rankings are inferred:

IDENT(low⋁high) ≫ *[+low]/V[−stressed, μ] ≫ IDENT(low)

IDENT(low⋁high) ≫ *[+low]/V[−stressed, μ] ≫ *[−high]/V[−stressed, μ] ≫ IDENT(high)

Of these, the ranking relationships between IDENT(low) and *[−high]/V[−stressed, μ], as well as between IDENT(low) and IDENT(high), remain undetermined. The reasons are as follows:

The winning candidates supported by IDENT(low) — namely, i e ɛ ɔ o u → i ɨ ɨ u u u — are all [+high], and thus also supported by the markedness constraint *[−high]/V[−stressed, μ]. As a result, these two constraints do not conflict.

As for IDENT(low) and IDENT(high), there are cases where the winning candidate is favoured by IDENT(low), and others where it is favoured by IDENT(high). In either case, however, the candidate is also supported by a higher-ranked constraint, and thus no conflict between IDENT(low) and IDENT(high) becomes decisive in the evaluation.

Now from (45), the descriptive generalisation (48) can be formulated:

(48) In unstressed syllables, short vowels that stand in opposition with respect to [*ATR*] are prohibited. This requirement is enforced by the unfaithful mapping [±*ATR*] → [0 *ATR*], as in *morada* "adress" /mɔr-ad-a/ → mɔˈra.da → m**u**ˈɾa.dɐ vs. *mɔˈɾa.dɐ, *m**o**ˈɾa.dɐ, *m**ʊ**ˈɾa.dɐ.

From this description, the following constraint hierarchies can also be posited:

***[±*ATR*]/V[−stressed, μ] ≫ IDENT(*ATR*)**

The definitions of the above constraints are as follows:

***[±*ATR*]/V[−stressed, μ]**: Assign one * for each unstressed short vowel that are specified with respect to [*ATR*].

IDENT(*ATR*): Assign one * for each change in the [*ATR*] feature between input and output.

The following tableaux (49) and (50) serve to confirm the ranking proposed above (note that the contextual specification /V[−stressed, μ] is omitted due to space constraints):

(49)

a	ID(low∨high)	*[+low]	*[−high]	ID(high)	*[+low]	ID(low)	*[±*ATR*]	ID(*ATR*)
☞ɐ			*			*		*
a		*W	*		*W	L		*
ɜ			*			*	*W	L
ɨ	*W		L	*W		*		*

(50)

ɔ	ID(low∨high)	*[+low]	*[−high]	ID(high)	*[+low]	ID(low)	*[±*ATR*]	ID(*ATR*)
☞u				*				*
ɔ			*W	L			*W	L
o			*W	L			*W	*
ʊ				*			*W	L

Finally, from (46), the descriptive generalisation (51) can be formulated:

(51) In unstressed syllables, front short vowels specified as [−back] are prohibited. This requirement is enforced by an unfaithful mapping from [−back] to [+back] — for example, *dedada* "fingermark" /d**e**d-ad-a/ → d**e**ˈda.da → d**ɨ**ˈdadɐ vs. *d**e**dada, *d**i**ˈdadɐ, and *pregar* "to nail" /prɛg-a-r0/ → pr**ɛ**ˈga.ɾ → pr**ɨ**ˈga.ɾ vs. *pr**ɛ**ˈga.ɾ, *preˈga.ɾ, *pr**i**ˈga.ɾ.

From this generalisation, the following ranking can be inferred:

***[−back]/V[−stressed, μ] ≫ IDENT(back)**

The definitions of the above newly introduced constraints are as follows:

***[−back]/V[−stressed, μ]**: Assign one * for each unstressed front short vowels.

IDENT(back): Assign one * for each change in the [back] feature between input and output.

The following tableau (52) serves to confirm the ranking proposed just above:

(52)

e	*[−back]/V[−stressed, μ]	IDENT(back)
☞ɨ		*
i	*W	L

The mapping of the winner vs. the loser is as follows:

/e/:[−low]$\boxed{\text{[−high]}}$[+*ATR*]$\boxed{\textbf{[−back]}}$→[ɨ]:[−low]$\boxed{\text{[+high]}}$[+*ATR*]$\boxed{\textbf{[+back]}}$

/e/:[−low]$\boxed{\text{[−high]}}$[+*ATR*]$\boxed{\textbf{[−back]}}$→*[i]:[−low]$\boxed{\text{[+high]}}$[+*ATR*]$\boxed{\textbf{[−back]}}$

However, **the mapping [−back] → [+back] always occurs in tandem with the mapping**

[−high] → [+high], and never in isolation — that is, /e ɛ/ → [ɨ] vs. *[i], whereas /i/ → [i] vs. *[ɨ]. Therefore, if we evaluate the output candidates [i] and *[ɨ] for /i/ based on the previously established ranking *[−back]/V[−stressed, μ] ≫ IDENT(back), *[ɨ], which is favoured by the higher-ranked *[−back]/V[−stressed, μ], would incorrectly win over [i], which is favoured by the lower-ranked IDENT(back), as shown in the tableau (53). This contradicts the observed facts:

(53)

i	*[−back]/V[−stressed, μ]	IDENT(back)
i	*	W
☞ɨ	L	*

If we reverse, however, the ranking between *[−back]/V[−stressed, μ] and IDENT(back), adopting IDENT(back) ≫ *[−back], we can no longer account for the mapping /e/ → [ɨ] vs. *[i]. Thus, **to derive the correct outputs**, it is necessary **to maintain the ranking *[−back/V [−stressed, μ] ≫ IDENT(back), while positing an additional constraint ranked above *[−back]/V[−stressed, μ] that favours /i/ → [i] and disfavors /i/ → *[ɨ]**.

Among the output candidates for /i/, [i] and *[ɨ] differ only in the value of the [back] feature — the former being [−back] and the latter [+back] — while they share identical values for the remaining features: [+high], [−low], [+*ATR*], and [−round]. Therefore, no markedness constraint pertaining to those shared features, even if ranked above *[−back], can favour [i] over *[ɨ]: such constraints would support neither candidate. In fact, among all the markedness constraints extracted thus far, only *[−back]/V[−stressed, μ] expresses any preference between [i] and *[ɨ]; the remaining constraints — *[+low]/V[−stressed, μ], *[−high]/V[−stressed, μ], and *[−*ATR*]/V [−stressed, μ] — favour neither. Even the introduction of additional markedness constraints such as *[−round] would yield the same result.

It is reasonable, thus, to assume that **the constraint ranked above *[−back] must be some faithfulness constraint**. Until its specific content is determined, we shall refer to this constraint provisionally as F. When this ranking — **F ≫ *[−back]/V[−stressed, μ] ≫ IDENT(back)** — is incorporated into the tableau discussed above, it yields the following structure (54) and (55) (note that no * is assigned in the column for F, as its definition remains unspecified):

(54)

e	F	*[−back]/V[−stressed, μ]	IDENT(back)
☞ɨ			*
i	(W)	*W	L

(55)	i	F	*[−back]/V[−stressed, μ]	IDENT(back)
	☞i		*	W
	ɨ	W	L	*

The faithfulness constraint F must satisfy the following two conditions within the ranking F ≫ *[back]/V[−stressed, μ] ≫ IDENT(back):

(56) F must **favour the mapping /i/ → [i] while disallowing /i/ → *[ɨ]**.

(57) F must not disallow the mapping /e/ → [ɨ]; that is, it must either **be neutral with respect to both /e/ → [ɨ] and /e/ → *[i]**, or else actively **favour /e/ → [ɨ]**.

Let us first consider the faithfulness violations involved in the mapping of /i/, with respect to condition (56). The mapping /i/ → [i] incurs no violations, whereas /i/ → *[ɨ] involves a single faithfulness violation: the change from [−back] to [+back]. Therefore, the faithfulness constraint F can be taken as one that **prohibits the mapping [−back] → [+back], thereby disfavoring /i/ → *[ɨ]**. However, since this constraint must outrank *[−back], it is evidently not identical to IDENT(back) **—** given that *[−back] ≫ IDENT(back).

Next, let us examine the faithfulness violations involved in the mappings of /e/, with respect to condition (57). The mapping /e/ → *[i] incurs a single violation: [−high] → [+high]. On the other hand, /e/ → [ɨ] incurs two violations: [−high] → [+high] and [−back] → [+back]. That is, **/e/ → [ɨ] entails more faithfulness violations than /e/ → *[i]**. It is therefore **unlikely that a faithfulness constraint F could favour /e/ → [ɨ] while disallowing /e/ → *[i]**. Consequently, the most plausible assumption is that **the faithfulness constraint F supports neither /e/ → [ɨ] nor /e/ → *[i]**.

Moreover, if the faithfulness constraint F were to target only the mapping [−back] → [+back] in /e/ → [ɨ], and prohibit it, then F would effectively be IDENT(back), which would contradict our assumption that F must outrank *[−back] while IDENT(back) does not. Therefore, **the faithfulness constraint F must also be concerned with the shared mapping [−high] → [+high] found in both /e/ → [ɨ] and /e/ → *[i], and likely serves to prohibit that change as well**.

To summarise the above discussion, **the faithfulness constraint F in question must simultaneously prohibit both the change [−back] → [+back] and [−high] → [+high]**. In order for such a constraint to penalise only the latter in the mapping /i/ → *[ɨ], while assigning the same number of violations to both /e/ → *[i] and /e/ → [ɨ], we can define F as **IDENT(high∧back)**. The definition of this constraint is as follows:

IDENT(high∧back): Assigns one violation mark * to a segment if there is a change in the value of either [high] or [back].

By ranking it as follows:

IDENT(high∧back) ≫ *[−back]/V[−stressed, μ] ≫ IDENT(back)

we can coherently account for the mappings /i/ → [i] vs. *[ɨ] and /e/ → [ɨ] vs. *[i] without contradiction. Put simply, within the [−low] domain defined by the [±high] and [±back] features, any deviation — whether a single step (as in /i/ → [ɨ] or /e ɛ/ → [i]) or a double step (as in /e ɛ/ → [i] → [ɨ]) — results in an equal degree of violation. And when the same violations are incurred, the candidate that achieves a more optimal shape — namely, /e ɛ/ → [i] → [ɨ] — emerges as the winner. The following tableaux (58) and (59) serve to confirm the ranking inferred above:

(58)

e	IDENT(high∧back)	*[−back]/V[−stressed, μ]	IDENT(back)
☞ɨ	*		*
i	*	*W	L

(59)

i	IDENT(high∧back)	*[−back]/V[−stressed, μ]	IDENT(back)
☞i		*	
ɨ	*W	L	*W

The introduction of the complex constraint IDENT(high∧back) here serves not merely as a technical expedient, but as **a theoretical expression of a functional requirement of the system**: **the preservation of the underlying distinctive opposition between high front /i/ and mid front vowels /e ɛ/**. Thus, the ranking of IDENT(high∧back) ≫ [−back]/V[−stressed, μ] should be interpreted as reflecting the system's resistance to neutralising this distinctive opposition. Without such a constraint, the system would favour maximal reduction to a reduced vowel [ɨ] across the entire front vowel space /i e ɛ/, thereby collapsing a opposition that is both frequent and functionally load-bearing.

While the general vowel reduction pattern in unstressed syllables preserves the distinctive opposition between /i/ and /e ɛ/ through the ranking IDENT(high∧back) ≫ [−back]/V[−stressed, μ], a more neutralised pattern is observed in final prosodically weakest syllables — namely, in final unstressed open syllables or in final unstressed syllables closed by [ʃ] or [ʒ], the only consonants specified as [−sonorant] in syllable coda — where even /i/ surfaces as [ɨ], as shown in (60):

(60) *part*[íː]*mos* ≈ *part*[ɨ]; *com*[éː]*mos* ≈ *com*[ɨ]; *diss*[ɛ́ː]*mos* ≈ *diss*[ɨ]

(Note that *partimos* and *parte* are the first person plural and third person singular of the verb *partir* "to depart" in the present indicative; *comemos* and *come* are the first person plural and third person singular of *comer* "to eat" in the present indicative; *dissemos* and *disse* are the first person plural and third person singular of *dizer* "to say" in the imperfect subjunctive.) This neutralisation may be accounted for by a positional markedness constraint *[−back]/V[−stressed, μ]__([−sonorant])#, which dominates IDENT(high∧back) specifically in these environments. The definition of this constrait is as follows:

***[−back]/V[−stressed, μ]__([−sonorant])#**: Assign one * for each unstressed short front vowel that occurs in a final open syllable or in a final syllable closed by a [−sonorant] segment.

The meaning of this constraint is that, in the weakest unstressed position at word-final edge (cf. Mattoso Câmara 1972), the underlying opposition between /i/ and /e ɛ/ can no longer be maintained. In Brazilian Portuguese as well, in this same weak unstressed position, the opposition between /i/ and /e ɛ/ is likewise neutralised, yielding /i e ɛ a ɔ o u/ → [i i i ɐ u u u]; by contrast, in other unstressed positions, the opposition is preserved, as seen in /i e ɛ a ɔ o u/ → [i e e a o o u].

Now, since IDENT(high∧back) is a faithfulness constraint that applies not only to [back] but also to [high], it comes into conflict not only with markedness constraints applying to the former — such as [−back]/V[−stressed, μ] and [back]/V[−stressed, μ]_([−sonorant])# — but also with the markedness constraint [−high]/V[−stressed, μ], which applies to the latter. For example, in the mapping e → ɨ vs. e̹ — e.g. *dedada* /ded-ad-a/ → deˈda.da → dɨˈda.dɐ vs. de̹ˈda.dɐ — the following patterns of constraint satisfaction and violation are observed:

e → ɨ: obeys $\boxed{\text{*[−high]/V[−stressed, μ]}}$, *[±*ATR*]/V[−stressed, μ], *[−back]/V[−stressed, μ];

violates IDENT(high), IDENT(*ATR*), $\boxed{\text{IDENT(high∧back)}}$, Ident(back)

e → *e̹:: obeys IDENT(high), *[±*ATR*]/V[−stressed, μ], $\boxed{\text{IDENT(high∧back)}}$, Ident(back);

violates $\boxed{\text{*[−high] /V[−stressed, μ]}}$, IDENT(*ATR*), *[−back]/V[−stressed, μ]

This suggests the ranking:

***[−high] /V[−stressed, μ] ≫ IDENT(high∧back)**

Moreover, according to the descriptive generalisation (47), *[−high]/V[−stressed, μ] ≫ IDENT(high), and according to (58) and (59), IDENT(high∧back) ≫ *[−back]/V[−stressed, μ] ≫ IDENT(back). Therefore, the presence or absence of violations of IDENT(high), [−back]/V[−stressed, μ] and IDENT(back) has no effect on this ranking.

Now let us turn to the problem of the long vowels. In this stratum, the long vowels [ɛː aː ɔː] undergo no change in either quality or quantity. This suggests that the presumed inviorable constraint **IDENT-QUANTITY** is at work, and that **the faithfulness constraints IDENT(high), IDENT(low), IDENT(*ATR*), IDENT(back) and IDENT(round) dominate all the markedness constraints applying to the features [high], [low], [*ATR*], [back] and [round] — with the exception of *[+low]/V[−stressed, μ], *[−high]/V [−stressed, μ], *[−back]/V [−stressed, μ] and *[±*ATR*]/V[−stressed, μ]**. The definition of the above presumed inviorable constraint is as follows:

IDENT-QUANTITY: Assign one * per mora difference between input and output — e.g. a difference of 1 mora incurs 1 *, a difference of 2 mora incurs 2 *, etc.

To summarise, in order to derive the mapping i e ɛ a ɔ o u ɛː aː ɔː → i ɨ ɨ ɐ u u u ɛː aː ɔː, it suffices to have the following constraint interactions:

(61) IDENT-QUANTITY;

(62) IDENT(low∨high) ≫ *[+low]/V[−stressed, μ] ≫ *[−high]/V[−stressed, μ] ≫ IDENT (high)

(63) *[+low]/V[−stressed, μ] ≫ IDENT(low)

(64) *[±*ATR*]/V[−stressed, μ] ≫ IDENT(*ATR*)

(65) *[−back]/V[−stressed, μ]__([−sonorant])# ≫ *[−high]/V[−stressed, μ]

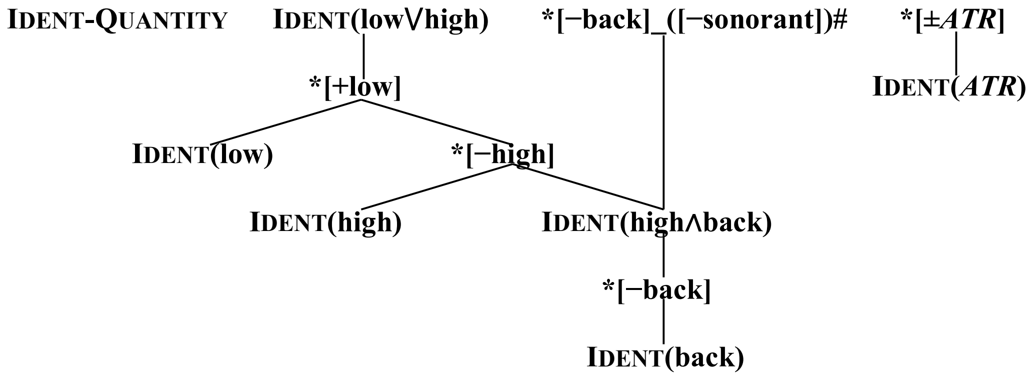
≫ IDENT(high∧back) ≫ *[−back]/V [−stressed, μ] ≫ IDENT(back)

(66) IDENT(high), IDENT(low), IDENT(*ATR*), IDENT(round) ≫ *[+high], *[−high], etc.

Among these, IDENT-QUANTITY, IDENT(low∨high), *[±*ATR*]/V[−stressed, μ], *[−back]/V

[−stressed, μ]__([−sonorant]) are assumed to be inviorable constraits.

The Hasse diagrams for the constraint hierarchies in (61)–(66) are shown in (67), where the context specification /V[−stressed, μ] is omitted for clarity:

(67) 

This was not intended at all, but it is noteworthy that the constraint hierarchy *[+low]/V [−stressed, μ] ≫ *[−high]/V[−stressed, μ] ≫ *[−back]/V[−stressed, μ] in (67) coincides with the process of vowel weakening in unstressed syllables analysed in (36)–(38), which suggests that the constraint hierarchy reflects the functioning of the system.

The following tableaux (68) - (74) serve to test the constraint ranking established in (67):

(68) *bolacha* /bol-aʃ-a/

boˈla.ʃa	☞buˈla.ʃɐ	buˈla.ʃa	buˈla.ʃɨ	boˈla.ʃɐ
IDENT-QUANTITY				
IDENT(low⋁high)	W		*	
*[+low]/V[−stressed, μ]	W	*		
*[−high]/V[−stressed, μ]	*	*	L	**
IDENT(high)	*	*	**	L

*[+low]/V[−stressed, μ]	W	*		
IDENT(low)	*	L	*	*

*[±*ATR*]/V[−stressed, μ]	W			*
IDENT(*ATR*)	**	**	**	*L

*[−back]/V[−stressed, μ]_([−sonorant])#				
IDENT(low⋁high)	W		*	
*[+low]/V[−stressed, μ]	W	*		
*[−high]/V[−stressed, μ]	*	*	L	**
IDENT(high⋀back)	*	*	**	L
*[−back]/V[−stressed, μ]				
IDENT(back)				

(69) *morada* /mɔr-ad-a/

mɔˈra.da	☞m**u**ˈɾa.dɐ	m**ɔ**ˈɾa.dɐ	m**o**ˈɾa.dɐ	m**ʊ**ˈɾa.dɐ
IDENT-QUANTITY				
IDENT(low⋁high)				
*[+low]/V[−stressed, μ]				
*[−high]/V[−stressed, μ]	*W	**	**	*
IDENT(high)	*	L	L	*

*[+low]/V[−stressed, μ]				
IDENT(low)	*	*	*	*

*[±*ATR*]/V[−stressed, μ]	W	*	*	*
IDENT(*ATR*)	*	L	*	L

*[−back]/V[−stressed, μ]_([−sonorant])#				
*[+low]/V[−stressed, μ]				
*[−high]/V[−stressed, μ]	*W	**	**	*
IDENT(high⋀back)	*	L	L	*
*[−back]/V[−stressed, μ]				
IDENT(back)				

(70) *pr**e**gar* /prɛg-a-r0/

pɾɛˈga.ɾ̩	☞pɾ**ɨ**ˈga.ɾ̩	pr**ɛ**ˈga.ɾ̩	pɾ**e**ˈga.ɾ̩	pɾ**i**ˈga.ɾ̩
IDENT-QUANTITY				
IDENT(low⋁high)				
*[+low]/V[−stressed, μ]				
*[−high]/V[−stressed, μ]	W	*	*	
IDENT(high)	*	L	L	*

*[+low]/V[−stressed, μ]				
IDENT(low)				

*[±*ATR*]/V[−stressed, μ]	W	*	*	
IDENT(*ATR*)	*	L	*	*

*[−back]/V[−stressed, μ]_([−sonorant])#				
IDENT(low⋁high)				
*[+low]/V[−stressed, μ]				
*[−high]/V[−stressed, μ]	W	*	*	
IDENT(high⋀back)	*	L	L	*
*[−back]/V[−stressed, μ]	W	*	*	*
IDENT(back)	*	L	L	L

(71) *c**i**nema* /s**i**nem-a/

s**i**ˈne.ma	☞s**i**ˈnemɐ	s**ɨ**ˈne.mɐ
IDENT-QUANTITY		
IDENT(lowVhigh)		
*[+low]/V[−stressed, μ]		
*[−high]/V[−stressed, μ]	*	*
IDENT(high)		

*[+low]/V[−stressed, μ]		
IDENT(low)	*	*

*[±*ATR*]/V[−stressed, μ]		
IDENT(*ATR*)	**	**

*[−back]/V[−stressed, μ]_([−sonorant])#		
IDENT(lowVhigh)		
*[+low]/V[−stressed, μ]		
*[−high]/V[−stressed, μ]	*	*
IDENT(high∧back)	W	*
*[−back]/V[−stressed, μ]	*	L
IDENT(back)	W	*

(72) *part**e*** /part-**i**/

ˈpaɾ.t**i**	☞ˈpaɾ.t**ɨ**	ˈpaɾ.t**i**
IDENT-QUANTITY		
IDENT(lowVhigh)		
*[+low]/V[−stressed, μ]		
*[−high]/V[−stressed, μ]		
IDENT(high)		

*[+low]/V[−stressed, μ]		
IDENT(low)		

*[±*ATR*]/V[−stressed, μ]		
IDENT(*ATR*)	*	*

*[−back]/V[−stressed, μ]_([−sonorant])#	W	*
IDENT(lowVhigh)		
*[+low]/V[−stressed, μ]		
*[−high]/V[−stressed, μ]		
IDENT(high∧back)	*	L
*[−back]/V[−stressed, μ]	W	*
IDENT(back)	*	L

(73) *come* /kɔm-**e**/

ˈkɔ.m**e**	☞ˈkɔ.m**ɨ**	ˈkɔ.m**e**	ˈkɔ.m**i**
IDENT-QUANTITY			
IDENT(low∨high)			
*[+low]/V[−stressed, μ]			
*[−high]/V[−stressed, μ]	W	*	
IDENT(high)	*	L	*

*[+low]/V[−stressed, μ]			
IDENT(low)			

*[±*ATR*]/V[−stressed, μ]	W	*	
IDENT(*ATR*)	*	L	*

*[−back]/V[−stressed, μ]_([−sonorant])#	W	*	*
IDENT(low∨high)			
*[+low]/V[−stressed, μ]			
*[−high]/V[−stressed, μ]	W	*	
IDENT(high∧back)	*	L	*
*[−back]/V[−stressed, μ]	W	*	*
IDENT(back)	*	L	

(74) *disse* /dis-**ɛ**/

ˈdi.s**ɛ**	☞ˈdi.s**ɨ**	ˈdi.s**ɛ**	ˈdi.s**e**	ˈdi.s**i**
IDENT-QUANTITY				
IDENT(low∨high)				
*[+low]/V[−stressed, μ]				
*[−high]/V[−stressed, μ]	W	*	*	
IDENT(high)	*	L	L	*

*[+low]/V[−stressed, μ]				
IDENT(low)				

*[±*ATR*]/V[−stressed, μ]	W	*	*	
IDENT(*ATR*)	*	L	*	*

*[−back]/V[−stressed, μ]_([−sonorant])#	W	*	*	*
IDENT(low∨high)				
*[+low]/V[−stressed, μ]				
*[−high]/V[−stressed, μ]	W	*	*	
IDENT(high∧back)	*	L	L	*
*[−back]/V[−stressed, μ]	W	*	*	*
IDENT(back)	*	L	L	L

4.2. Postlexical Stratum 2: Vocalic Quantity Adjustment

From (23), it was assumed that major vowel quality changes occur at Postlexical Stratum 1 in (27), and that only quantity adjustment takes place in this stratum as shown in (28). Accordingly, in this stratum, IDENT-QUALITY is assumed to be an inviolable constraint at work. The definition of this constraint is as follows:

IDENT-QUALITY: Assign one * for each change in any quality feature — e.g. [low], [high], [*ATR*], [back], [round] — between Input and Output.

Turning to vowel quantity alternations in Contemporary EP, the following descriptive generalisation may be formulated as (75):

(75) Long vowels are prohibited. This requirement is enforced by deleting one mora if the input is a long vowel (e.g. *ativo* "active" /aːt-iv-o/ → aːˈti.vu → aˈtiː.vu, *aːˈtiː.vu), suggesting the ranking ***μμ/V ≫ MAX(μ)**,

except when the vowel occurs in a stressed syllable, where short vowels are prohibited. This requirement is enforced by epenthesising one mora if the input is a short vowel (e.g. *ativo* "active" /aːt-iv-o/ → aːˈti.vu → aˈtiː.vu vs. *aˈti.vu; *pasta* "file-folder" /past-a/ → ˈpaʃ.tɐ → ˈpaːʃ.tɐ vs. *ˈpaʃ.tɐ), suggesting the ranking ***ˈσ[μ] ≫ DEP(μ)**, or by maintainig the length if the input is a long vowel (e.g. *ata* "minutes, record" /aːt-a/ → ˈaː.tɐ → ˈaː.tɐ vs. *ˈa.tɐ; *mestre* "master" /mɛːstr-e/ → ˈmɛːʃ.tɾɨ → ˈmɛːʃ.tɾɨ vs. *ˈmɛʃ.tɾɨ), suggesting the ranking ***ˈσ[μ] ≫ *μμ/V**,

except when the vowel occurs in a closed syllable ending in [ɫ], [ɾ], [i̯] or [u̯], where long vowels are prohibited. This requirement is enforced by deleting one mora if the input is a long vowel (e.g. *norma* "regulation, norm" /nɔːrm-a/ → ˈnɔːɾ.mɐ → ˈnɔɾ.mɐ vs. *ˈnɔːɾ.mɐ; *normal* "normal" /nɔːrm-al-0/ → nɔːɾˈma.l̩ → nɔɾˈmaː.l̩ vs. *nɔːɾˈmaː.l̩), suggesting the ranking ***μμμ ≫ *ˈσ[μ]**.

From the descriptive generalisation (75), the constraint rankings (76) and (77) may be inferred:

(76) *μμμ ≫ *ˈσ[μ] ≫ *μμ/V ≫ MAX(μ)

(77) *μμμ ≫ *ˈσ[μ] ≫ *μμ/V, DEP(μ)

The definitions of the above constraints are as follows:

***μμμ**: Assign one * for each syllable with three moras.

***ˈσ[μ]**: Assign one * for each stressed syllable with only one mora.

***μμ/V**: Assign one * for each vowel linked to two moras.

MAX(μ): Assign one * for each mora deleted.

DEP(μ): Assign one * for each mora epenthesised.

On the other hand, syllable quantity adjustment can also be achieved not through moraic operations but by the epenthesis or deletion of segments. For example, in mappings such as *ata* (noun) /aːta/ → ˈaː.tɐ → ˈaː.tɐ vs. *ˈai̯.tɐ, or *norma* /nɔːrma/ → ˈnɔːɾ.mɐ → ˈnɔɾ.mɐ vs. *ˈnɔː.mɐ, the asterisked candidates *ˈai̯.tɐ and *ˈnɔː.mɐ are eliminated due to violations of DEP and MAX, respectively. The definitions of these constraints are as follows ("segment" here refers to a root node in the quality feature geometry abstracting away from quantity; therefore, deletion or insertion of moras alone is not considered a violation of DEP or MAX):

DEP: Assign one * for each segment epenthesised.

MAX: Assign one * for each segment deleted.

First, in the mapping *ata* ˈaː.tɐ → ˈaː.tɐ vs. *ˈai̯.tɐ, the winner ˈaː.tɐ is favoured by DEP, while the loser *ˈai̯.tɐ is favoured by *μμ/V, which indicates the ranking **DEP ≫ *μμ/V**. Note that both candidates violate none of the constraints *μμμ, *ˈσ[μ], MAX, MAX(μ), or DEP(μ), and thus are equally evaluated with respect to these constraints. Next, in the mapping /aːt-a/ → ˈaː.tɐ → ˈaː.tɐ vs. *tɐ, the winning candidate ˈaː.tɐ is favoured by MAX, whereas the losing candidate *tɐ is favoured by *μμ/V, which supports the ranking **MAX ≫ *μμ/V**. Note also that ˈaː.tɐ satisfies MAX(μ), while *[tɐ] violates it; however, since *μμ/V ≫ MAX(μ), candidate evaluation by MAX(μ) does not affect the evaluation determined by *μμ/V. Since MAX and DEP do not conflict in syllable quantity adjustment, the following constraint ranking (78) can be derived from the above discussion:

(78) MAX, DEP >> *μμ/V

Next is the ranking among MAX, DEP and *μμμ, all of which are ranked above *μμ/V. With the exception of word-final trimoraic syllables — such as [-ˈmeːɾ] in *comer* /kɔm-e-r0/ → [kuˈmeːɾ] "to eat" — which require special theoretical treatment and which shall be discussed below, trimoraic syllables are found only in loanwords, as in *e-mail* "e-mail"/imeil/ → iˈmɐi̯ɫ, with the final syllable -ˈmɐi̯ɫ. In such cases, neither epenthesis — as in *i.mɐi̯.lɨ — nor deletion — as in *iˈmɐi̯ — is attested, which suggests, at least provisionally, the ranking (79):

(79) Max, Dep ≫ *μμμ

As stated in (75), short vowels in stressed CV syllables surface as long vowels by the addition of one mora. However, when such an open syllable is immediately followed by a vowel-initial syllable — that is, when the stressed vowel and the following unstressed vowel form a hiatus — the situation becomes somewhat complex. The short vowels /i/, /u/, and /o/ undergo lengthening without any change in quality: *via* "road" /vi-a/ → ˈvi.ɐ → [ˈviː.ɐ]; *rua* "street" /ru-a/ → ˈʀu.ɐ → [ˈʀuː.ɐ]; *boa* "good" /bo-a/ → ˈbo.ɐ → [ˈboː.ɐ]. The short vowels /e/ and /ɛ/, specified as [−low, −high, −back], on the other hand, undergo lengthening and concomitant diphthongisation in i̯ or epenthesis of i̯ immediately following them: *veia* "vein" /ve-a/ → ˈve.a → ˈve.ɐ → ˈvei̯.ɐ; *ateia* "atheist" /a-tɛ-a/ → aˈtɛ.a → ɐˈtɛ.ɐ → ɐˈtɛi̯.ɐ. (Subsequently, the dissimilation ei̯, ɛi̯ → [ɐi̯] in a later stratum — here tentatively called Postlexical Stratum 3 or Surface-phonetic Stratum: the phonetic-phonological interface — produces the final surface forms *veia* [ˈvɐi̯.ɐ] and *ateia* [ɐˈtɐi̯.ɐ] in EP, whereas in BP this dissimilation does not take place and ei̯ and ɛi̯ surface as [ei̯] and [ɛi̯], as in *veia* [ˈvei̯.ɐ] and *ateia* [aˈtɛi̯.ɐ]. In this stratum, a ranking such as the tentatively named OCP(back)/Rhyme — a constraint prohibiting adjacent [αback] (where α is a variable) within the same rhyme, as in *[ei̯], *[ɛi̯], *[ou̯] — ranked above Dep(back) and Max(back) might account for the exclusion of ill-formed outputs in EP such as *[ˈvei̯.ɐ], *[ɐˈtɛi̯.ɐ], and *douto* /dout-o/ "erudite" *[ˈdou̯.tu] for example.)

The next issue is how to analyse the change é.ɐ → éi̯.ɐ and ɛ́.ɐ → ɛ́i̯.ɐ. It is possible, as was suggested above, to interpret this change not as the epenthesis of a segmental i between e or ɛ and ɐ, but rather as the result of lengthening e and ɛ by one mora, during which the features [+high] and [0*ATR*], are added sequentially after the original [−high] and [±ATR] specifications within the internal feature geometry of e and ɛ, as shown in (80), where the added feature values are shaded:

(80)

	μ e		μμ ei̯	
low	−		−	
high	−		−	+
ATR	+	⇒	+	0
back	−		−	
round	−		−	

	μ ɛ		μμ ɛi̯	
low	−		−	
high	−		−	+
ATR	−	⇒	−	0
back	−		−	
round	−		−	

This anlysis is conceptually analogous to the geometry analysis of the affrication of plosives, where [+continuant] is added sequentially at the root node after [−continuant]. From the above dicussion, the following descriptive generalisation (81) may be formulated:

(81) In a stressed open syllable before another syllable beginnig by a vowel, short vowels are prohibited. This requirement is enforced by simply epenthesising one mora if the input is a short vowel (e.g. *via* /vi-a/ → ˈvi.ɐ → ˈviː.ɐ, *vi.ɐ; *rua* /ru-a/ → ˈʀu.ɐ → ˈʀuː.ɐ, *ˈʀu.ɐ; *boa* /bo-a/ → ˈbo.ɐ → ˈboː.ɐ, *bo.ɐ) suggesting the ranking ***ˈσ[μ] ≫ DEP(μ)** as in (75),

except when a front mid long vowel — namely, éː or ɛ́ː in this language — would result, in which case [+high], or [+high] and [+ATR], are added sequentially after the original [−high] and [−ATR] within the feature geometry of e and ɛ (e.g. *vea* /ve-a/ → ˈve.a → ˈve.ɐ → ˈvei̯.ɐ, *ˈve.ɐ, *veː.ɐ; *ateia* /a-tɛ-a/ → aˈtɛ.a → ɐˈtɛ.ɐ → ɐˈtɛi̯.ɐ, *ɐˈtɛ.ɐ, *ɐˈtɛː.ɐ), suggesting the rankings ***LONGMIDFRONT_.V ≫ DEP(high), DEP(*ATR*)** and ***ˈσ[μ] ≫ DEP(high), DEP(*ATR*)**,

furthermore, the off-glide corresponding to the epenthesised mora is prohibited from being [−high, ±ATR] (e.g. *ateia* /a-tɛ-a/ → aˈtɛ.a → ɐˈtɛ.ɐ → ɐˈtɛi̯.ɐ, *ɐˈtɛe̯.ɐ, *ɐˈtɛɪ̯.ɐ), suggesting the ranking ***NON-HIGH-OFFGLIDE ≫ DEP(high), DEP(ATR)**.

From the descriptive generalisation (81), the constraint rankings (82) - (84) may be inferred:

(82) *LONGMIDFRONT_.V ≫ DEP(high), DEP(*ATR*)

(83) *ˈσ[μ] ≫ DEP(μ), DEP(high), DEP(*ATR*)

(84) *NON-HIGH-OFFGLIDE ≫ DEP(high), DEP(*ATR*)

The definitions of the above constraints are as follows:

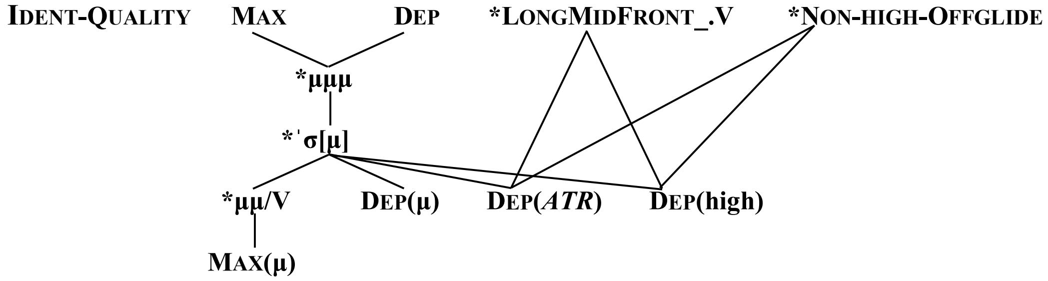
***LONGMIDFRONT_.V**: Assign one * for each front mid long vowel occurring before another vowel in another syllable.

DEP(high): Assign one * for each instance of [high] that is newly inserted into the original feature geometry of a segment.

DEP(*ATR*): Assign one * for each instance of [*ATR*] that is newly inserted into the original feature geometry of a segment.

***NON-HIGH-OFFGLIDE**: Assign one * for each vocoid specified either as [−high] or as [±ATR] that occurs as an off-glide within the same nucleus.

The Hasse diagrams for the constraint hierarchies in (76) - (84) are as shown in (85):

(85) 

Let us return to the trimoraic [-ˈmeːɾ] in *comer* [kuˈmeːɾ]. Assuming that the unmarked stress assignment in EP — based on binary foot construction starting from the word-final syllable — also applies to impersonal infinitives such as *falar*, *comer* and *partir*, the underlying representation of *comer* is most likely /kɔm-e-r0/, where /0/ denotes a vowel with no segmental quality but with a vocalic quantity of one mora, as stated in (26) (cf. Section 6.6). When syllabified, this underlying representation is assumed to take the form kɔ.me.r̩, with the consonant /r/ linked to the floating mora of /0/, which at Lexical Stratum 3 receives main stress on the syllable -me-, yielding the derived form kɔˈme.r̩. This representation, having undergone the lenition of r̩ to ɾ̩ at some point between stress assignment and vowel nasalisation, yields at Postlexical Stratum 1 the form kuˈme.ɾ̩, which subsequently undergoes quantity adjustment at Postlexical Stratum 2 to produce the output kuˈmeː.ɾ̩. The process is summarised in (86):

(86) *comer* /kɔm-e-r0/ → kɔ.me.r̩ → kɔˈme.r̩ → kuˈme.ɾ̩ → kuˈmeː.ɾ̩

cf. *comera* /kɔm-e-ra/ → kɔ.me.ra → kɔˈme.ra → kuˈme.ɾɐ → kuˈmeː.ɾɐ

Subsequently, resyllabification in a later stratum — tentatively called Postlexical Stratum 3 or Surface-phonetic Stratum — produces the final surface form [kuˈmeːɾ]. In this stratum, the ranking *C̩, MAX(μ), MAX, DEP ≫ *μμμ accounts for the exclusion of ill-formed outputs such as *[kuˈmeɾ], *[kuˈmeː.ɾɨ], and *[kuˈmeː]. Under this analysis, the output of /kɔm-e-r0/ at this stratum is kuˈmeː.ɾ̩, which does not violate the constraint *μμμ.

The following tableaux (87) – (93) serve to test the constraint ranking established as in (85):

(87) *ativo* /aːt-iv-o/

aːˈti.vu	ID-QUAL	MAX	DEP	*μμμ	*ˈσ[μ]	*μμ/V	MAX(μ)
☞aˈtiː.vu						*	*
aːˈtiː.vu						**W	L
aˈti.vu					*W	L	*
aːˈti.vu					*W	*	L

*ˈσ[μ]	DEP(μ)
	*
	*
*W	L
*W	L

(88) *past* /pasta/

ˈpas.tɐ	ID-QUAL	MAX	DEP	*μμμ	*ˈσ[μ]	*μμ/V	MAX(μ)
☞ˈpaːʃ.tɐ						*	
ˈpaʃ.tɐ					*W	L	

*ˈσ[μ]	DEP(μ)
	*
*W	L

(89) *ata* /aːt-a/ (noun)

ˈaː.tɐ	ID-QUAL	MAX	DEP	*μμμ	*ˈσ[μ]	*μμ/V	MAX(μ)
☞ˈaː.tɐ						*	
ˈa.tɐ					*W	L	*W
ˈai̯.tɐ			*W			L	
tɐ		*W				L	**W

*ˈσ[μ]	DEP(μ)
*W	

(90) *mestre* /mɛːstr-e/

ˈmɛːʃ.tɾɨ	ID-QUAL	MAX	DEP	*μμμ	*ˈσ[μ]	*μμ/V	MAX(μ)
☞ˈmɛːʃ.tɾɨ						*	
ˈmɛʃ.tɾɨ					*W	L	*W

*ˈσ[μ]	DEP(μ)
*W	

(91) *norma* /nɔːrm-a/

ˈnɔːɾ.mɐ	ID-QUAL	MAX	DEP	*μμμ	*ˈσ[μ]	*μμ/V	MAX(μ)
☞ˈnɔɾ.mɐ							*
ˈnɔːɾ.mɐ				*W		*W	L
ˈnɔː.mɐ		*W				*W	*

*ˈσ[μ]	DEP(μ)

(92) *normal* /nɔːrm-al-0/

nɔːrˈma.l̩	ID-QUAL	MAX	DEP	*μμμ	*ˈσ[μ]	*μμ/V	MAX(μ)
☞nɔɾˈmaː.l̩						*	*
nɔːɾˈmaː.l̩				*W		**W	L

*ˈσ[μ]	DEP(μ)
	*
	*

(93) *comer* /kɔm-e-r0/

kuˈme.ɾ̩	ID-QUAL	MAX	DEP	*μμμ	*ˈσ[μ]	*μμ/V	MAX(μ)
☞kuˈmeː.ɾ̩						*	
kuˈme.ɾ̩					*W	L	
kuˈmeː		*W				*	*W
kuˈmeː.ɾi			*W			*	
kuˈmeːɾ				*W		*	

*ˈσ[μ]	DEP(μ)
	*
*W	L
	*
	**W
	*

In Contemporary EP, the constraint ranking outlined above ensures that underlying long vowels retain their length faithfully only in stressed open syllables and in stressed closed syllables whose coda is [ʃ] or [ʒ]. In these contexts, furthermore, underlying short vowels are also realised as long vowels due to the ranking *ˈσ[μ] ≫ *μμ/V ≫ MAX(μ), which results in a surface-level neutralisation of vowel length opposition. The only context where this opposition could potentially remain is in unstressed syllables; even there, however, *μμ/V ≫ MAX(μ) enforces surface-level neutralisation of length. In such cases, the underlying opposition is preserved through vowel quality reduction: only underlying short vowels undergo weakening in vowel quality. Thus, since **the underlying opposition in vowel length is either neutralised on the surface or reinterpreted as a quality opposition**, it is therefore likely that **native speakers of EP do not perceive the opposition in vowel length as of distinctive nature**.

5. Analysis summary

6. Theoretical applications

In this section, we shall examine several phenomena that can be accounted for by appealing to (i) the opposition between underlying short vowels /$\overset{\mu}{\text{i}}\ \overset{\mu}{\text{e}}\ \overset{\mu}{\text{ɛ}}\ \overset{\mu}{\text{a}}\ \overset{\mu}{\text{ɔ}}\ \overset{\mu}{\text{o}}\ \overset{\mu}{\text{u}}$/ and underlying long vowels /$\overset{\mu\mu}{\text{ɛ}}\ \overset{\mu\mu}{\text{a}}\ \overset{\mu\mu}{\text{ɔ}}$/, as posited in Section 3.2., and (ii) the postlexical phonological processes discussed in Section 4, namely vowel quality reduction and vowel quantity adjustment, both of which are sensitive to stress and to the context in which the vowel is found and take place across the two strata of the postlexical level.

6.1. Lack of quality reduction of nasal vowels in unstressed syllables

Contemporary EP typically exhibits five distinct nasal monophthongs [ĩ ẽ ɐ̃ õ ũ] at the phonetic level[9]. These vowels undergo quantity alternation — [ĩ́ː ẽ́ː ɐ̃́ː ṍː ṹː] ≈ [ĩ ẽ ɐ̃ õ ũ] — between stressed and unstressed syllables, but typically do not undergo the kind of vowel quality reduction characteristic of underlying short oral vowels, as seen in *p*[ɛ́ː]*sa* ≈ *p*[ɨ]*samos* from the verb *pesar* "to weigh" and *m*[ɔ́ː]*ra* ≈ *m*[u]*ramos* from the verb *morar* "to live". For example, the verb *pensar* "to think" shows the alternation *p*[ẽ́ː]*sa* ≈ *p*[ẽ]*samos*, but not *p*[ẽ́ː]*sa* ≈ **p*[ɨ̃]*samos*; similarly, the verb *contar* "to tell, to count" exhibits the alternation *c*[ṍː]*ta* ≈ *c*[õ]*tamos*, but not

[9] In certain boundaries between clitics or between clitics and words, one may find another nasal vowel [ã] — distinct from [ɐ̃] — as in <u>*a an*</u>*tiga* [ãˈtiːɣa] — which will be discussed later in this chapter.

c[ṍː]*ta* ≈ **c*[ũ]*tamos*. Why, then, do these vowels not undergo quality reduction? **The fact that, in alternations between stressed and unstressed nasal vowels, only vowel quantity changes while quality remains constant is a property shared with the underlying long vowels** — /ɛː aː ɔː/ → [ɛ́ː áː ɔ́ː] ≈ [ɛ a ɔ]. Does this, then, mean that nasal vowels are underlyingly long vowels?

When discussing nasal vowels, the initial problem lies in the phonological interpretation of the nasal vowel itself. In the case of Portuguese, there exist two competing analyses: one views the nasal vowel as *a single segment*, while the other considers it as *a phonetic realisation of a sequence of two segments*. The former is represented by works such as Lüdke (1953), but is no longer a widely accepted view today. The latter, by contrast, branches into two theoretical traditions: (i) a functionalist–structuralist interpretation, according to which the nasal vowel is the phonetic realisation of a sequence composed of a vocalic phoneme and an archiphoneme /ɴ/ (Barbosa 1994, etc.); and (ii) an autosegmental account, in which nasal vowels derive phonologically from a sequence consisting of a vocalic segment and an autosegment [+nasal] (Mateus and Andrade 2002; Mateus et al. 2003, etc.). This study aligns with the latter two *two-segment sequence* analyses, and argues — on the basis of the three considerations presented below — that the phonetically realised simple nasal vowel [Ṽ] in Portuguese is phonologically /Vn/; in other words, a sequence consisting of a vocalic segment /V/ followed by a nasal consonantal segment /n/.

The first piece of evidence lies in the phonological alternation [Vn] ≈ [Ṽ] observed within the same morpheme under different phonological environments. For instance, the alternation [i.n-] ≈ [ĩ.-] of the negative prefix *in-* in *in+ativo* [i.naˈtiːvu] "inactive" vs. *in+sanável* [ĩ.sɐˈnaːvɛɫ] "incurable" — where + marks the boundary between a derivational prefix and the base — and the alternation [iɾ.mɐˈn-] ≈ [iɾ.mɐ̃.-] of the root morpheme *irman-* in *irman+ar* [iɾ.mɐˈnaːɾ] "to join together" vs. *irman+dade* [iɾ.mɐ̃ˈdaːðɨ] "brotherhood" suggest the following: an underlying sequence /Vn/ is syllabified as {V}{nV} before another vowel /V/, whereas before a consonant-vowel sequence /CV/ it is syllabified as {Vn}{CV}, where { } indicates syllabic parsing. In the former case {V}{nV}, the underlying /n/ is syllabified as the onset of the following syllable and is phonetically realised as segmental [n]. In the latter {Vn}{CV}, /n/ is syllabified as the coda of the preceding syllable, nasalises the preceding vowel /V/, and is then deleted at the phonetic level — {Vn}{CV} → {Ṽn}{CV} → {Ṽ~~n~~}{CV} → {Ṽː}{CV}. As illustrated by the alternation *cannot* [ˈkænɑːt] ≈ *can't* [ˈkæ̃ːnt] in American English, it is a cross-linguistically common phenomenon for a vowel to become nasalised before a nasal coda. Moreover, as seen in certain dialects of American English where *can't* is realised as [kæ̃ːt], the nasal coda [n] may be deleted after nasalising the preceding vowel (cf. Kenstowicz 1994: 14). These observations suggest that the interpretation proposed above is not phonologically implausible.

The second piece of evidence concerns the phonetic realisation of the consonant /r/. This segment is realised as a tap or a flap [ɾ] when it follows an oral vowel, i.e. /VrV/ → [VɾV] — as in *caro* "expensive" /kar-o/ → [ˈkaːɾu] — but typically as a trill [ʀ] (or a fricative [ʁ], or more conservatively as a trill [r]) when it follows another consonant such as /s/, /l/, or /r/, i.e. /VsrV/, /VlrV/, /VrrV/ → [VʒʀV], [VɫʀV], [VʀV] — as in *Israel* /israɛl/ → [iʒʀɐˈɛɫ], *melro* "blackbird" /mɛlro/ → [ˈmɛɫʀu], and *carro* "car" /karro/ → [ˈkaːʁu]. (In /VrrV/, the first /r/ is assumed to be deleted following the change of the second /r/ to [ʀ], probably due to some OCP-related constraint.) This distribution shows that when /r/ occurs in syllable-initial position, it is realised as [ɾ] if preceded by a [−consonantal] segment, and as [ʀ] if preceded by a [+consonantal] one. Now, since /r/ is also realised as [ʀ], but not as [ɾ], following a nasal vowel — e.g. [ṼʀV] as in *genro* "son-in-law" [ˈʒẽːʀu] — it is reasonable to assume that, phonologically, a [+consonantal] segment intervenes between the vowel and /r/. That is, **the sequence [ṼʀV] should be analysed phonologically as /VnrV/**.

The third piece of evidence concerns the phonetic realisation of the consonants /b d g/. In Contemporary EP, it is generally the case that in the phonic environment [V_V], these segments assimilate the redundant feature [+continuant] of the preceding vowel and are realised as fricatives [β ð ɣ] — e.g. /VbV/, /VdV/, /VgV/ → [VβV], [VðV], [VɣV], as in *o boi* [uˈβoi̯] "the ox", *o dia* [uˈðiːɐ] "the day", *o gato* [uˈɣaːtu] "the cat". By contrast, in the environment [Ṽ_V], these same consonants are constantly realised as stops [b d g] — e.g. [Ṽ]/bV/, [Ṽ]/dV/, [Ṽ]/gV/ → [ṼbV], [ṼdV], [ṼgV], as in *um boi* [ũˈboi̯] "an ox", *um dia* [ũˈdiːɐ] "a day", *um gato* [ũˈgaːtu] "a cat".

This alternation /b d g/ → [β ð ɣ] ≈ [b d g] after oral versus nasal vowels supports the analysis that [Ṽ_V] is phonologically /Vn_V/: since the consonantal /n/, unlike vowels, is specified as [−continuant], the following /b d g/ cannot assimilate [+continuant] from it, and their underlying [−continuant] specification is preserved in surface realisation. (For example, *longe* /lonʒe/ "far (away)" is phonetically realised as [ˈlõːʒɨ], and not as *[ˈlõː**d**ʒɨ]; this indicates that the consonant following /n/ — in this case, /ʒ/, which is [+continuant] — does not assimilate the [−continuant] feature of /n/; it follows, then, that /b d g/ are underlyingly [−continuant] stops, since they cannot assimilate the [−continuant] feature of /n/.) From these three points, the interpretation that "the phonetically nasalised vowel [Ṽ] is derived from the underlying segmental sequence /Vn/" may be regarded as phonologically well grounded.

From the above discussion, two main points (94) and (95) have emerged:

(94) Nasal vowels exhibit the same type of alternation as underlying long vowels. That is, in alternations between stressed and unstressed syllables, vowel quantity changes but vowel quality remains constant — namely, [ĩ́: ẽ́: ɐ̃́: ṍ: ṹ:] ≈ [ĩ ẽ ɐ̃ õ ũ], just as the underlying long vowels /ɛ: a: ɔ:/ → [ɛ́: á: ɔ́:] ≈ [ɛ a ɔ].

(95) Nasal vowels are the phonetic realisation of a segmental sequence consisting of a vocalic segment followed by a nasal consonant segment, i.e., /Vn/.

What can be inferred from (94) and (95) is the following: **nasal vowels, at the input to Postlexical Strata 1 and 2, where vowel quality modification and vowel quantity adjustment take place respectively, are derived long vowels of bimoraic weight —** namely, **ĩ: ẽ: ɐ̃: õ: ũ: —** just like the underlying long vowels **—** ɛ: a: ɔ:. As long vowels, they are subject to the same constraint hierarchy and receive the same treatment as underlying long vowels with respect to both quality and quantity. Precisely because they are bimoraic, they **undergo no change in quality at the vowel quality modification stratum** — just like underlying long vowels — and then, **at the vowel quantity adjustment stratum**, they are **realised with full quantity in stressed syllables, but lose one mora in unstressed syllables**, undergoing shortening in the output — **ĩ́: ẽ́: ɐ̃́: ṍ: ṹ: ≈ ĩ: ẽ: ɐ̃: õ: ũ: → ĩ́: ẽ́: ɐ̃́: ṍ: ṹ: ≈ ĩ ẽ ɐ̃ õ ũ**, just as ɛ́: á: ɔ́: ≈ ɛ: a: ɔ: → ɛ́: á: ɔ́: ≈ ɛ a ɔ. On the other hand, since they are underlyingly segmental sequences /Vn/, **nasal vocalisation of the /Vn/ sequence — that is, the nasalisation of /V/ and the subsequent deletion of /n/ — must occur** prior to the vowel quality modification stratum — that is, **between the syllabification stratum and the quality modification stratum**; furthermore, insofar as they surface as long vowels, the /Vn/ sequence must be bimoraic, and **no mora deletion takes place during the process of nasal vocalisation**, as shown in (96):

(96)

$\overset{\mu\ \mu}{\text{/Vn/}}$		$\overset{\mu\mu}{\text{/V/}}$		Input representation
in en ɛn an ɔn on un	in en ɛn an ɔn on un	ɛ: a: ɔ:	ɛ: a: ɔ:	**Syllabification**
ín én ɛ́n án ɔ́n ón ún	—	ɛ́: á: ó:	—	**Stress Assignment**
ĩ́: ẽ́: ẽ́: ɐ̃́: ṍ: ṍ: ṹ:	ĩ: ẽ: ẽ: ɐ̃: õ: õ: ũ:	—	—	**Vowel Nasalisation**
—	—	—	—	**Vowel Quality Modification**
—	ĩ ẽ ẽ ɐ̃ õ õ ũ		ɛ a ɔ	**Vowel Quantity Adjustment**
ĩ́: ẽ́: ẽ́: ɐ̃́: ṍ: ṍ: ṹ:	ĩ ẽ ẽ ɐ̃ õ õ ũ	ɛ́: á: ó:	ɛ a ɔ	Output representation

At the vowel nasalisation stratum, the mapping **en ɛn an ɔn on un → ĩ: ẽ: ɐ̃: õ: õ: ũ:** takes place. In particular, the mapping **ɛn an ɔn → ẽ: ɐ̃: õ:** — where **raising of the tongue-height is attested — resembles the quality reduction of unstressed underlying short vowels /e ɛ a ɔ o/ → [ɨ ɨ ɐ u u]**, and can likewise be analysed as **a kind of reduction**. Because expiratory airflow is partially consumed by nasal resonance, the volume of airflow reaching the oral cavity is reduced, which results in the neutralisation of vowel quality oppositions.

Relevant to this interpretation that vowel nasalisation in Portuguese is a form of vocalic reduction, the adverb ***on**tem* "yesterday" may be pronounced idiolectically as [ˈ**ɔ̃**:tɐ̃ĩ̯], with open-mid [ɔ̃], instead of [ˈ**õ**:tɐ̃ĩ̯], with close-mid [õ] — the latter form being the phonetic representation canonically registered in dictionaries — as Barbosa (1994: 99) noted: "Não faltam em português … variantes individuais da realização de fonemas. Muitas passam despercebidas à maioria dos utentes, como é o caso … da realização como [ọ̃] ou [ǫ̃] da vogal acentuada de *ontem*", which would correspond to [õ] and [ɔ̃] in IPA, respectively. This word originates from the Latin phrase *ad noctem* or *hāc nocte* and, according to Williams (1975: 94), underwent the following changes: *hac nŏcte* > **a nocte* > *aõite* > *oonte* … > *ontem*. From the etymological point of view, therefore, the *o-* in *ontem* derives from the vowel sequence *oo-* in *oonte* and is assumed to be an underlying long vowel /$\overset{\mu\mu}{\text{ɔ}}$/: accordingly, the underlying representation of this adverb is /$\overset{\mu\mu}{\text{ɔ}}$nten/. This underlying long vowel /$\overset{\mu\mu}{\text{ɔ}}$/ — which, in unstressed syllables, undergoes reduction through the loss of one mora while retaining its open-mid quality and is realised as a short vowel [ɔ] — is in parallel assumed, in syllables closed by /n/, to undergo reduction through the loss of one mora while retaining its open-mid quality, to be nasalised as ɔ̃ and, with the subsequent deletion of /n/ and by compensatory lengthening, to become the derived long nasal vowel ɔ̃:, as shown in (97):

(97) /$\overset{\mu\mu}{\text{ɔ}}$nten/ → ɔ:n.ten → $\boxed{\text{ˈɔ:n.ten → ˈɔ̃n.tẽn}}$ → ˈɔ̃:.tẽ: (→ ˈɔ̃:.tɐ̃ĩ̯ → [ˈɔ̃:tɐ̃ĩ̯])

There also exists, on the other hand, another phonological representation /$\overset{\mu}{\text{ɔ}}$nten/ in which /$\overset{\mu\mu}{\text{ɔ}}$/ has been shortened and, in idiolects whose lexicon stores this representation, the underlying short vowel /$\overset{\mu}{\text{ɔ}}$/ — which, in unstressed syllables, undergoes reduction through the quality modification and is realised as a reduced vowel [u] — is in parallel assumed, in syllables closed by /n/, to undergo reduction through the quality modification from open-mid ɔ to close-mid o, to be nasalised as õ and, with the subsequent deletion of /n/ and by compensatory lengthening, to become the derived long nasal vowel õ:, as shown in (98):

(98) /$\overset{\mu}{\text{ɔ}}$nten/ → ɔn.ten → $\boxed{\text{ˈ\textbf{ɔ}n.ten → ˈ\textbf{õ}n.tẽn}}$ → ˈõː.tẽː (→ ˈõː.tẽi̯ → [ˈõːtɐ̃i̯])

Under this interpretation, also in the nasalisation of /V/ in the /Vn/ sequence, the general rule of reduction applies: *underlying long vowels lose quantity, while underlying short vowels lose quality*. Additional evidence in support of this generalisation is found in the adjective *antiga* "old" /$\overset{\mu}{\text{a}}$ntig-a/ → [ɐ̃ˈtiːɣɐ], where the sequence /$\overset{\mu}{\text{a}}$n-/, with an underlying short vowel /$\overset{\mu}{\text{a}}$/, is realised as the qualitatively reduced nasal vowel [ɐ̃-]. In contrast, in the phrase *a antiga* "the old" /$\overset{\mu}{\text{a}}$#$\overset{\mu}{\text{a}}$ntig-a/ → [ãˈtiːɣɐ] , the initial sequence *a an-*, containing a derived long vowel $\overset{\mu\mu}{\text{a}}$, is realised as the nasal vowel [ã-], which has undergone a loss of quantity but not of quality.

Note finally that expected constraint ranking in the mapping (90) are likely to be as follows:

IDENT-QUANTITY

*[−*ATR*]/__N]σ ≫ IDENT(*ATR*)

*[+low]/__N]σ ≫ IDENT(low)

IDENT[low⋁*ATR*]

*VN ≫ MAX(C)

The constraint hierarchy might be as follows:

IDENT[low⋁*ATR*] ≫ *[+low]/__N]σ ≫ *[−*ATR*]/__N]σ ≫ IDENT(*ATR*)

*[+low]/__N]σ ≫ IDENT(low)

*VN ≫ MAX(C)

6.2. Diphthongisation of vowel segments before palatal consonants

In Contemporary EP, as Barbosa (1988: 346) points out, "hoje em dia, se encontra com frequência um *iod* entre qualquer vogal acentuada e qualquer uma das quatro consoantes palatais indicadas inicial da sílaba seguinte, e isso quer tal *iod* possa quer não possa explicar-se etimologicamente, quer tenha quer não tenha representação gráfica: palavras como *caixa*, *graxa*, *olhos*, *hoje*, *unha*, *beijo*, *velho* são pronunciadas com ou sem [i̯], mais ou menos nítido, e isto varia de idiolecto para idiolecto e, por vezes, no mesmo idiolecto". The terms *as quatro consoantes palatais* and *iod* refer, in IPA terms, to [ʃ ʒ ʎ ɲ] and [i̯], respectively. In orthographic terms, these are represented as *x*, *ch* for [ʃ]; *j*, *g* (before *i*, *e*) for [ʒ]; *lh* for [ʎ]; and *nh* for [ɲ].

The glide [i̯], when graphically represented, appears as *i*.

For instance, in the third-person singular forms of the verbs *relaxar* "to relax" and *taxar* "to tax" in the present indicative, the stressed vowel, spelled *a* in both cases, exhibits variation between a monophthong and a diphthong ending in [i̯]: *rel*[áː]*xa* ≈ *rel*[ái̯]*xa* and *t*[áː]*xa* ≈ *t*[ái̯]*xa*. What is rather interesting is the realisation of this radical vowel *a* in unstressed syllables. In the former case, the same type of variation between a monophthong and a diphthong — *rel*[a]*xar* ≈ *rel*[ai̯]*xar* — is again observed in the unstressed syllable, just as in the stressed one. In contrast, in the latter case, only the monophthong is attested to our knowledge: *t*[ɐ]*xar*. What does this imply? (The following is a reinterpretation of Makino (2006[1]: 25-29) within the theoretical framework adopted in this study.)

The alternation between the stressed and unstressed vowels without [i̯] in these verbs — namely, *rel*[áː]*xa* ≈ *rel*[a]*xar* and *t*[áː]*xa* ≈ *t*[ɐ]*xar* — coincides with the alternation of an underlying long vowel /$\overset{\mu\mu}{a}$/ in the former, and that of an underlying short vowel /$\overset{\mu}{a}$/ in the latter, as indicated in (23). Based on this, we assume the underlying representations of these verbs to be ***relaxar*** /rel$\overset{\mu\mu}{\mathbf{a}}$ʃ-a-r0/ and ***taxar*** /t$\overset{\mu}{\mathbf{a}}$ʃ-a-r0/, respectively.

Next, let us turn to the behaviour of [i̯]. The variation or alternation between a monophthong — here abbreviated as V — and a diphthong — abbreviated as Vi̯ — is limited to open syllables preceding the prepalatal and palatal consonants [ʃ ʒ ʎ ɲ], and does not occur before [s z l n], which are adjacent to them in relation to the place of articulation feature within the consonant system. It is therefore clear that the [i̯] element in Vi̯ is in some way conditioned by articulatory properties that are present in [ʃ ʒ ʎ ɲ] but absent in [s z l n]. From a diachronic perspective, many of these consonants appear to have arisen through coalescence involving mutual assimilation between coronal consonants such as [s z l n] and the dorsal glide [j] or [i̯]. Examples include lat. *rŭssĕu-* > pt. *ro*[ʃ]*o*; lat. *cervēsĭa-* > pt. *cerve*[ʒ]*a*; lat. *sĕdĕa* > pt. *se*[ʒ]*a*; lat. *allĭu-* > pt. *a*[ʎ]*o*, lat. *tĕnĕō* > te[ɲ]o (cf. Williams 1961: 42, 44, 46, 50, 91). A similar process is observed in contemporary English, as in *I miss you* [ʌɪ̯ mɪs ju], which can phonetically surface as [ʌɪ̯ mɪʃu] due to the coalescence of [-s j-] into [-ʃ-] via mutual assimilation.

In Contemporary EP, the underlying segments /s z ʎ ɲ/ are generally interpreted as simple segments with a single place of articulation. However, in the present study — taking into account both their historical development and their articulatory characteristics in the contemporary language — these palatal consonants are analysed as complex segments involving two distinct oral constrictions: a primary articulation formed by the tongue crown against the alveolar ridge — a gesture shared with /s z l n/, represented as the node **Coronal-[+anterior]** in feature geometry (cf. Kenstowicz 1994: 136–168); and a secondary articulation formed by the tongue front against the hard palate — a gesture characteristic of [j i̯], represented as the node **Dorsal-[−back, +high]**. If phonological features are understood as neural commands issued by the brain to the speech organs in order to execute specific muscular gestures (cf. Kenstowicz 1994: 20, 138), then in the case of /ʃ ʒ ʎ ɲ/, one may say that two distinct constriction commands co-occur at the phonological level: a command to the tongue crown to constrict against the alveolar ridge, and a command to the front part of the tongue body to constrict against the hard palate. These co-occurring commands give rise, at the phonetic level, to articulations that range from post-alveolar to pre-palatal to palatal (for the place of articulation of [ʃ ʒ ʎ ɲ], cf. Barbosa 1994[1]: 63).

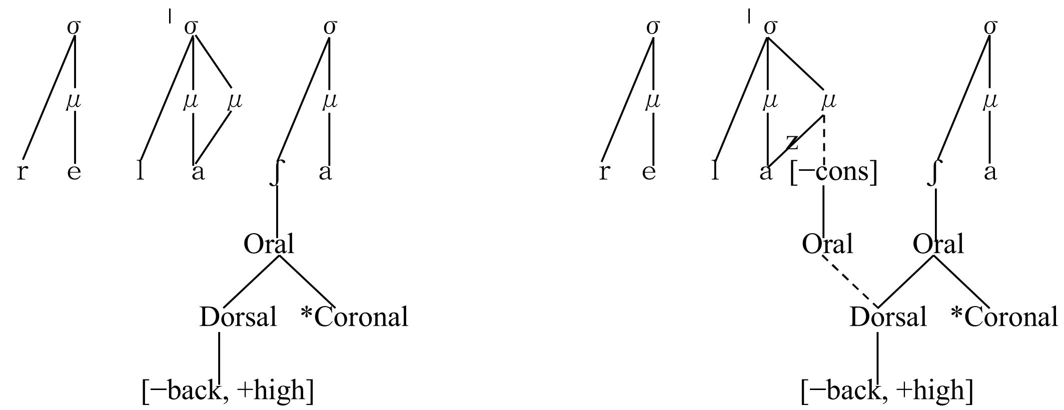
The above discussion can be summarised in the following two points:

(99) The underlying representations of the verbs *relaxar* and *taxar* are /rel$\overset{\mu\mu}{a}$ʃ-a-r0/ and /t$\overset{\mu}{a}$ʃ-a-r0/, respectively.

(100) The consonants /ʃ ʒ ʎ ɲ/ are analysed as complex segments involving two simultaneous oral gestures: a coronal constriction at the alveolar ridge — corresponding to the node Coronal-[+anterior] — and a dorsal constriction at the hard palate — corresponding to the node Dorsal-[−back, +high].

With the assumptions in (99) and (100), it becomes possible to account for the variation observed in *rel*[áː]*xa* ≈ *rel*[ái̯]*xa* ≈ *rel*[a]*xar* ≈ *rel*[ai̯]*xar* and *t*[áː]*xa* ≈ *t*[ái̯]*xa* ≈ *t*[ɐ]*xar*. Let us begin with the mapping /rel$\overset{\mu\mu}{a}$ʃ-a/ ≈ /rel$\overset{\mu\mu}{a}$ʃ-a-r0/ → *rel*[áː]*xa* ≈ *rel*[a]*xar*. Here, the stressed vowel [áː] is selected as the output corresponding to the input underlying long vowel /$\overset{\mu\mu}{a}$/, evaluated as the optimal candidate under the constraint ranking in (85). The unstressed [a] is similarly derived. In contrast, in the mapping /rel$\overset{\mu\mu}{a}$ʃ-a/ ≈ /rel$\overset{\mu\mu}{a}$ʃ-a-r0/ → *rel*[ái̯]*xa* ≈ *rel*[ai̯]*xar*, the vowel quality /a/ of the underlying long vowel /$\overset{\mu\mu}{a}$/ is first delinked from the second mora, resulting in /$\overset{\mu}{a}$/. Then, to satisfy the vocalic constriction required by a syllable nucleus, a floating node [−consonantal, +sonorant]-Oral is introduced and linked to the delinked mora. Finally, the node Dorsal-[−back, +high] — the secondary palatal articulation inherent in /ʃ/ — spreads onto this node as in (101):

(101) [ʀɨˈlaːʃɐ] [ʀɨˈlai̯ʃɐ]



The resulting complex — /$\overset{\mu\mu}{a}$ʃ/ → $\overset{\mu}{a}\,\overset{\mu}{}$ʃ → $\overset{\mu\mu}{\text{ai̯}}$ʃ — is then evaluated under the constraint ranking in (85), and its optimal phonetic realisations emerge as [ái̯ʃ] and [ai̯ʃ] as in (102):

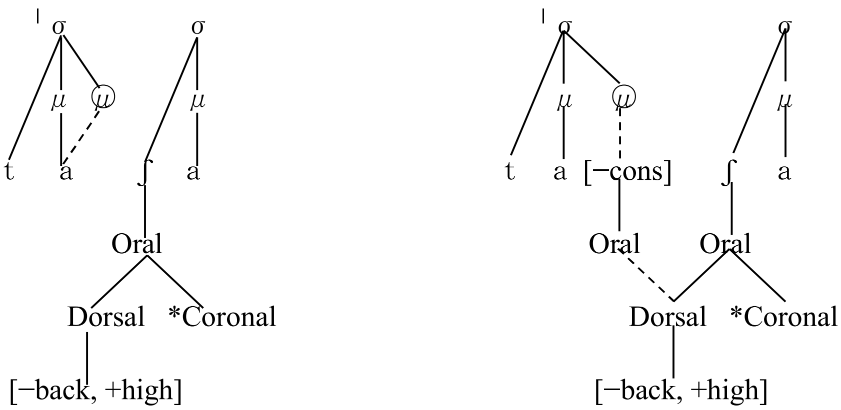
(102) *rel**a**xa* /rel$\overset{\mu\mu}{\mathbf{a}}$ʃ-a/

ʀɨˈlaːʃɐ	ID-QUAL	MAX	DEP	*μμμ	*ˈσ[μ]	*μμ/V	MAX(μ)
☞ʀɨˈlaːʃɐ						*	
☞ʀɨˈlai̯ʃɐ						*	
ʀɨˈlaɾʃɐ			*W			L	
ʀɨˈlaʃɐ					*W		*W

*ˈσ[μ]	DEP(μ)

Next, consider the mapping /t$\overset{\mu}{a}$ʃ-a/ → *t*[áː]*xa*. In this case, the vowel quality /a/, which is associated with only one mora in the underlying representation /$\overset{\mu}{a}$/, becomes associated with an additional mora — represented by ⓜ in (97 left) below — under stress in Postlexical Stratum 2, resulting in a long vowel [áː]. This long vowel is then selected as the output corresponding to the input underlying short vowel /$\overset{\mu}{a}$/, as it is evaluated to be the optimal candidate under the constraint ranking given in (85). On the other hand, in the mapping /t$\overset{\mu}{a}$ʃ-a/ → *t*[ái̯]*xa*, the vowel quality /a/ of the underlying short vowel /$\overset{\mu}{a}$/ is not linked to the additional mora assigned under stress and is realised as a short vowel [á]. In its place, an [−consonant, +sonorant]−Oral node — introduced to satisfy the syllable nucleus's requirement for vocalic constriction — is linked to that mora. Finally, the Dorsal-[−back, +high] node, corresponding to the secondary palatal articulation of /ʃ/, spreads onto this Oral node, as shown in (103 right):

(103) [ˈtaːʃɐ] [ˈtai̯ʃɐ]



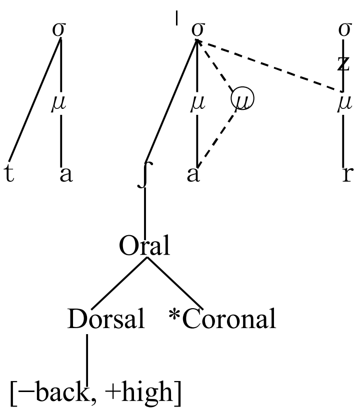
The resulting segment sequence is then selected as the optimal output — [ái̯ʃ] — under the constraint ranking given in (85). The candidate evaluation tableau in this case is given in (104):

(104) *taxa* /t$\overset{\mu}{\mathbf{a}}$ʃ-a/

ˈtaʃɐ	ID-QUAL	MAX	DEP	*μμμ	*ˈσ[μ]	*μμ/V	MAX(μ)	*ˈσ[μ]	DEP(μ)
☞ˈtaːʃɐ						*			*
☞ˈtai̯ʃɐ						*			*
ˈtaɾʃɐ			*W			L			*
ˈtaʃɐ					*W	L			L

Finally, in the mapping /t$\overset{\mu}{a}$ʃ-a-r0/ → *t*[ɐ]*xar*, the segment [ɐ], which is the output of the underlying short vowel /$\overset{\mu}{a}$/ in Postlexical Stratum 2 and serves as the input to Postlexical Stratum 3, occupies an unstressed syllable position. As such, it neither receives an additional mora nor associates with one, as indicated in (105):

(105) [tɐˈʃaːɾ]



It is, therefore, selected as the output under the constraint ranking (85), as shown in (106):

(106) ***ta**xar* /t$\overset{\mu}{\textbf{a}}$ʃ-a-r0/

tɐ'ʃa.ɾ	ID-QUAL	MAX	DEP	*μμμ	*'σ[μ]	*μμ/V	MAX(μ)		*'σ[μ]	DEP(μ)
☞tɐ'ʃaː.ɾ						*				*
tɐi̯'ʃaː.ɾ						*				**W
tɐ'ʃa.ɾ					*W	L			*W	L

It is worth noting that, regarding this palatal extension, one of the informants cited in Makino (2005) made the particularly intriguing observation that "after speaking for a long time and becoming tired, I tends to say *rel*[ai̯]*xar* instead of *rel*[a]*xar*". Phonetically, this phenomenon may be explained by a form of regressive assimilation that facilitates the articulatory transition from /$\overset{\mu\mu}{a}$/ or /$\overset{\mu}{a}$/ to /ʃ/.

6.3. Phonetic realisation of vowel coalescence in phrasal phonology

It was already noted in Section 3.2 that the [a] in the imperative sentence *Fica aqui* "Stay here" → *Fic*[a]*qui*, where the locative adverb *aqui* follows the second person singular imperative form *fica* of the verb *ficar* "stay", can be interpreted as the phonetic realisation of a derived long vowel $\overset{\mu\mu}{a}$ at the phrasal level. A similar phenomenon is also observed within the domain of the prosodic word:

(107) *Dav**a-a***. /dava#a/ → [ˈdaːv**a**] "He/she/you used to give it."

(108) *Dav**a-o***. /dava#u/ → [ˈdaːv**ɔ**] "He/she/you used to give it."

Example sentence (107) is of the same type as *Fic*[a]*qui*, and appears to derive as follows: *Dav*$\overset{\mu}{\textbf{a}}$ -$\overset{\mu}{\textbf{a}}$ → *Dav*$\overset{\mu}{\textbf{a}}$-$\overset{\mu}{\cancel{\textbf{a}}}$ → *Dav*$\overset{\mu\mu}{\textbf{a}}$'. In this process, the vowel quality of the second segment *a* is deleted due to the OCP, the remaining mora is relinked to the preceding *a*, and a derived long vowel $\overset{\mu\mu}{a}$ is formed, which is realised as [a] at the surface level.

Example (108) is slightly more complex: the [–round] and [+low, +back] features of the vowel /a/ are delinked from the Labial and Dorsal nodes, respectively, in the feature geometry, allowing the spreading of the [+round] and [–low, +back] features of the following /o/[10]. Then,

[10] The personal pronoun *o* [u] is a clitic lexically specified as unstressed, and since no alternation with a stressed vowel exists, its underlying values for [high] and [*ATR*] are indeterminate. Theoretically, it could be /u/, /o/ or /ɔ/. However, based on orthographic evidence that an underlying /u/ is consistently written as *u* even in word-final position, as in *ven**ú**s-io* ≈*ven**u**s-to* ≈ *Vén**u**s* and *on**u**sto* ≈ *ón**u**s*, whereas

/o/ itself is delinked from its mora at the root node level, and a segment with altered feature values is relinked to the vacated mora, forming the derived long vowel $\overset{\mu\mu}{\text{ɔ}}$, as shown in (109), where the features retained from the two original segments in this coalescence are shaded. This vowel is subsequently shortened by one mora under constraint hierarchy (85), surfacing as the optimal output [ɔ]:

(109)

	$\overset{\mu}{\text{/a/}}$	$\overset{\mu}{\text{/o/}}$
low	+	−
high	−	−
ATR	−	+
back	+	+
round	−	+

⇒

	$\overset{\mu\mu}{\text{ɔ}}$
low	−
high	−
ATR	−
back	+
round	+

It is assumed here that in this spreading, the feature [+back] is delinked from /a/, because in the assimilation pattern /ae/ → [ɛ(ː)], as common as /ao/ → [ɔ(ː)], /a/ delinks its [+back] feature and accepts the spreading of [−back] from /e/. Moreover, the changes /ao/ → [ɔ(ː)] and /ae/ → [ɛ(ː)] exhibit a symmetrical pattern.

The derivation of long $\overset{\mu\mu}{\text{a}}$ within a prosodic word is also observed in forms such as the contraction of the preposition *a* [ɐ] + the definite article *a* [ɐ] → *à* [a], and the contraction of the preposition *a* [ɐ] + the demonstrative *aquilo* [ɐˈkiːlu] → *àquilo* [aˈkiːlu]. The fact that *a* and *à*, or *aquilo* and *àquilo*, are phonetically distinguishable in EP is due to the survival of the vowel quantity opposition. In BP, by contrast, long vowels have been eliminated from the vowel system, and the opposition of vowel quantity has collapsed — in BP, for example, *pr*[ɨ]*gar* "to nail", with short $\overset{\mu}{\text{/ɛ/}}$, and *pr*[ɛ]*gar* "to preach", with long $\overset{\mu\mu}{\text{/ɛ/}}$, which are distinguished by vowel quantity in EP, are homophones *pr*[e]*gar* with /ɛ/ in BP. As a result, *a* and *à*, or *aquilo* and

the nominal/adjectival thematic marker (índice temático) *-o* is written as *o* and are unlikely to be derived from an underlying /u/ — for example, the underlying forms of *vau* "ford", *véu* "veil", and *feio* "ugly", *fio* "string" may be analysed as /va/ + -o, /vɛ/ + -o and /fe/ + -o, /fi/ + -o, respectively; when the vowel preceding -o bears stress, these forms undergo two different types of syllabification; the former group surfaces as monosyllabic words [{váu̯}], [{vɛ́u̯}], whereas the latter as disyllabic words [{fɐ́i̯}{u}], [{fíː}{u}], where { } indicates syllable boundaries; this difference in syllabification can be straightforwardly explained if the underlying form of -o is assumed to be /o/ or /ɔ/. If *-o* is analysed as /o/, the underlying representations of these forms are /va-o vɛ-o/ and /fe-o fi-o/, respectively; in other words, when the vowel preceding *-o* /o/ is more open than /o/, as in /a ɛ/, monosyllabification occurs; when it is equally or more close, as in /e i/, disyllabification occurs; the same type of explanation holds if /ɔ/ is assumed as the underlying form; if /u/ is assumed, however, the difference between monosyllabification and disyllabification can no longer be explained in terms of the relative aperture between the preceding vowels /a ɛ e i/ and *-o* /u/ — it seems thus relatively unlikely that the underlying form of the pronoun spelled as *o* is /u/.

àquilo, are not phonetically distinguishable. The inability to distinguish in speech the contracted form *à* (from *a* + *a*) from either the simple preposition *a* or the simple article *a* constitutes a significant communicative problem. It may well be one of the reasons why, in contemporary spoken BP, alternative prepositions such as *para* or *em* are often used in contexts where *a* would appear in writing.

6.4. The distinctive opposition [áː] ≈ [ɐ́ː] in *-amos* and *-ámos* in the first-person plural of the present and preterite indicatives

In stressed CV syllables preceding +nasal consonants [m n ɲ], /a/ is realised as [ɐ́ː], reflecting a change from [+low] to [−low] — e.g. *cama* "bed" /kama/ → [ˈkɐːmɐ]; *ano* "year" /ano/ → [ˈɐːnu]; *banho* "bath" /baɲo/ or /banʲo/ → [ˈbɐːɲu]. This raising may be regarded as a type of weakening, possibly resulting from airflow redistribution: nasal resonance in vowels, triggered by the spreading of the [+nasal] feature, diverts part of the pulmonic airstream into the nasal cavity, thereby reducing the amount of airflow available for oral articulation and rendering the [+low] region inaccessible.

This phonological context also allows, nevertheless, for occurrences of the [+low] vowel [áː], though in highly restricted environments. These include the noun *g*[áː]*nho* "gain, profit" and the rizotonic forms of the verb *g*[a]*nhar* "to gain" — namely, *g*[áː]*nho*, *g*[áː]*nhas*, *g*[áː]*nha*, *g*[áː]*nham*, *g*[áː]*nhe*, *g*[áː]*nhes*, *g*[áː]*nhem* — as well as the first person plural of the preterite indicative in *-ar* regular verbs, encoded by the ending *-ámos* [-áːmuʃ]. This final form stands in systematic contrast with the first person plural of present indicative ending *-amos* [-ɐ́ːmuʃ], as exemplified by *fal*[áː]*mos* "we spoke" vs. *fal*[ɐ́ː]*mos* "we speak", both derived from the verb *falar* "to speak".

Accordingly, even in the context preceding [+nasal] consonantal segments, the mid vowel [ɐ́ː] and the low vowel [áː] stand in a distinctive opposition. The vowel *a* in the root *ganh-* is historically a long vowel of bimoraic weight, as noted by Teyssier (1981: 42): "Temos, com efeito, *ga-anha* > *ganha* (verbo) e *ga-anho* > *ganho* (substantivo), nos quais o *a*, resultante da contracção, conservou até hoje no português europeu um timbre aberto [a], apesar da presença da consoante nasal seguinte, que, nas palavras que contêm um a singelo etimológico, sempre fechou esta vogal em [ä]" — that is, [ɐ] or [ə] in IPA. And given that it does not undergo quality reduction even in unstressed syllables, as in *g*[a]*nhar*, it is reasonable to assume that it still retains its bimoraic status in contemporary usage, reflecting the alternation schema /$\overset{\mu\mu}{a}$/ → [áː] ≈

[a] discussed in (23). This vowel length may thus be interpreted as blocking the weakening process that would otherwise occur before nasal consonants.

As previously noted in Makino (2007[1]: 3), "in the preterite indicative, each combination of 'indicative-nonpast-finished' and a specific 'person-number' set appears morphologically as an *amalgam* — e.g. *fale-**i***, *fala-**ste***, *falo-**u***, *falá-mos* (← *fala-* + -$^{+\mu}$***mos***), *fala-**stes***, *fala-**ram***" (translated from Japanese). On the notion of amalgam — *amalgame* in French — Martinet (1991: 102) writes: "deux signifiés qui coexistent dans un énoncé enchevêtrent leurs signifiants de telle façon qu'on ne saurait analyser le résultat en segments successifs". The first person plural ending of the preterite indicative **$\overset{\mu}{-}$*mos*** — that is, **-$^{+\mu}$*mos*** in the example above — is thus a morphological amalgam of [−non-assertive, −VP∈past, −finished, −assumptive] with [−non-participant, −addressee, +plural]. It **begins not with a segmental element but with a floating mora** /$\overset{\mu}{\ }$/, much like "the Hebrew definite prefix *ha*", which "has the phonological form *haX*, where *X* is an empty position that prefers to be occupied by the following consonant but when this is impossible then by the preceding vowel" (Kenstowicz 1994: 44–45). In contrast, the ***-mos*** of the present indicative encodes only theperson-numer features [−non-participant, −addressee, +plural] and therefore lacks such a floating mora /$\overset{\mu}{\ }$/. If one assumes then that the theme vowel *a* of the verbal stem — for example, the vowel *a* of the stem *fal**a**-* of the verb *falar* — becomes linked to this initial mora, yielding a bimoraic structure as in (110), the form *fal*[áː]*mos* should be analysed on a par with /g$\overset{\mu\mu}{\text{a}}$ɲo/ → *g*[áː]*nho*. By contrast, in (111), such mora linking does not occur due to the absence of a floating mora:

(110) preterite indicative: *fal*$\overset{\mu}{a}$*-* + $\overset{\mu}{-}$ *mos* → *fal*$\overset{\mu}{a}$*-* $\overset{\mu}{\ }$*mos* → *fal*$\overset{\mu\mu}{\boldsymbol{a}}$*-mos* → *fal*[**á**ː]*mos*

(111) present indicative: *fal*$\overset{\mu}{a}$*-* + *-mos* → *fal*$\overset{\mu}{\boldsymbol{a}}$*-mos* → *fal*[**ɐ́**ː]*mos*

As discussed in section 6.1, the tautosyllabic vowel-nasal sequence Vn undergoes nasalisation of V and deletion of /n/ — in en ɛn an ɔn on un → ĩː ẽː ẽː ɐ̃ː õː õː ũː — prior to the Vowel Quality Modification Tier, and the tongue-height raising of an → ɐ̃ː during this process can be attributed to the influence of [+nasal] from the nasal segment n. Similarly, as argued just above, the raising of short á to ɐ́ in stressed CV syllables preceding [m, n, ɲ] is likewise attributable to the [+nasal] feature from these consonants. Given that both the raising in a[+nasal]. → ɐ̃ː and the raising in á.[+nasal] → ɐ́.[+nasal] — where . denotes a syllable boundary — are to be triggered by the same feature [+nasal], it seems unlikely that they should occur at different tiers. If we therefore assume that the latter process also precedes the Vowel Quality Modification Tier, then the short vowel /$\overset{\mu}{\text{a}}$/ and the long vowel /$\overset{\mu\mu}{\text{a}}$/ that occur in stressed CV syllables before

[+nasal] consonants would be realised as ɐ́ and áː, respectively, at the input to that tier. At this stage, they are evaluated as optimal intermediate outputs by the set of constraints (67) operative in this stratum, and passed on as inputs to the subsequent Vowel Quantity Adjustment Tier. As shown in the tableaux in (112) and (113), they are then output as [ɐ́ː] and [áː], respectively:

(112) *falamos* /fal-$\overset{\mu}{\text{a}}$-mos/

fɐˈlɐmuʃ	ID-QUAL	MAX	DEP	*μμμ	*ˈσ[μ]	*μμ/V	MAX(μ)		*ˈσ[μ]	DEP(μ)
☞ fɐˈlɐːmuʃ						*				*
fɐˈlɐmuʃ					*W	L			*W	L
fɐˈlamuʃ	*W				*W	L			*W	L
fɐˈlɐi̯muʃ			*W							*

(113) *falámos* /fal-$\overset{\mu}{\text{a}}$-$^{\mu}$ mos

fɐˈlaːmuʃ	ID-QUAL	MAX	DEP	*μμμ	*ˈσ[μ]	*μμ/V	MAX(μ)		*ˈσ[μ]	DEP(μ)
☞ fɐˈlaːmuʃ						*				
fɐˈlamuʃ					*W	L	*W		*W	
fɐˈlɐːmuʃ	*W					*				*

In the preterite indicative of regular *-ar* verbs, the endings *-i*, *-ste*, *-u*, *-stes*, and *-ram* are all morphological amalgams encoding both the TAM feature bundle [−non-assertive, −VP∈past, −finished, −assumptive] and the person–number values [±non-participant, ±addressee, ±plural]. In contrast, the present indicative endings — *-o*, *-s*, *-Ø*, *-is*, and *-m* — encode only the person–number values [±non-participant, ±addressee, ±plural] and lack any temporal-aspectual specification. It is therefore highly implausible, from a systemic point of view, to assume that only the first person plural of the preterite should share its ending *-mos* with the present indicative. Such an assumption would disrupt the overall morphological and semantic parallelism of the system and thus seems theoretically unnatural.

Accordingly, the surface contrast between [ɐ́ː] and [áː] in stressed CV syllables preceding nasal consonants [m n ɲ] is best interpreted as the phonetic realisation of an derived opposition between $\overset{\mu}{\text{a}}$ and $\overset{\mu\mu}{\text{a}}$ — that is, between a monomoraic short vowel and a bimoraic long vowel at Lexical Stratum 1: Feature Establishment in (24). On this analysis, thus, stressed CV syllables preceding [+nasal] consonants [m n ɲ] constitute the only type of stressed syllable in which the opposition between the short vowel $\overset{\mu}{\text{a}}$ and the long vowel $\overset{\mu\mu}{\text{a}}$ does not undergo neutralisation — precisely because of the quality weakening affecting the short vowel.

On the other hand, the situation is different for regular verbs of the *-er* and *-ir* conjugations. The theme vowels *e* and *i* of these verbs are the underlying short vowels /$\overset{\mu}{\text{e}}$/ and /$\overset{\mu}{\text{i}}$/, respectively. When these vowels are linked to the floating mora /$\overset{\mu}{\ }$/ of the preterite indicative first person plural ending -$\overset{\mu}{\ }$mos, they form derived long vowels $\overset{\mu\mu}{\text{e}}$ and $\overset{\mu\mu}{\text{i}}$. In stressed CV syllables, however, the surface realisation of these derived long vowels is identical to that of their short counterparts — both are realised phonetically as [éː] and [íː], respectively, and thus the opposition is neutralised at the phonetic level. This interpretation is based on the following inference: as shown in (23), the underlying long vowels /$\overset{\mu\mu}{\text{ɛ}}$/, /$\overset{\mu\mu}{\text{a}}$/, and /$\overset{\mu\mu}{\text{ɔ}}$/ are mapped to [ɛ́ː], [áː], and [ɔ́ː], respectively, in stressed CV syllables and, If systemic consistency and parallelism are to be maintained, then the derived long vowels $\overset{\mu\mu}{\text{e}}$ and $\overset{\mu\mu}{\text{i}}$ must likewise be mapped to [éː] and [íː], which coincide with the mapping of the underlying short vowels /$\overset{\mu}{\text{e}}$/ and /$\overset{\mu}{\text{i}}$/ in the same prosodic environment, also illustrated in (23). This is illustrated in (114) and (115) by comparing the first person plural form *comemos* in the preterite indicative with the superficially identical form *comemos* in the present indicative of the verb *comer* "to eat".

(114) preterite indicative: *com$\overset{\mu}{\text{e}}$-* + *-$\overset{\mu}{\ }$mos* → *com$\overset{\mu}{\text{e}}$-$\overset{\mu}{\ }$mos* → *com$\overset{\mu\mu}{\textbf{e}}$-mos* → *com*[**é**ː]*mos*

(115) present indicative: *com$\overset{\mu}{\text{e}}$-* + *-mos* → *com$\overset{\mu}{\textbf{e}}$-mos* → *com*[**é**ː]*mos*

6.5. Cases requiring further consideration

(i) Semantic transparency of morphemes and their prosodic structure

In some cases, semantic transparency of morphemes and their prosodic structure appear to be related to whether a vowel undergoes reduction. In *televisão* "television" and *telefone* "telephone", the prefix *tele-* /tɛlɛ-/ meaning "remote" is realised with reduced vowels as [tɨlɨ-]. By contrast, in *telemóvel* "mobile phone", the same string /tɛlɛ-/ originates as a clipped form of *telefone*, and is realised with non-reduced vowels [tɛlɛ-]. While this semantic difference may play a role in the preservation of full vowel quality, it is also possible that, in *telemóvel*, the element *tele-* constitutes a prosodic word on its own, as a semantically transparent lexical constituent — unlike the *tele-* in *televisão* or *telefone*. In that case, the primary stress of the prosodic word would fall on [ˈtɛ-]. The remaining issue concerns the final unstressed vowel [-ɛ], but this may be treated analogously to the final unstressed [-ɔ] in *euro*, when pronounced as [ˈeu̯ɾɔ].

In Makino (2010), based on the observation that native speakers perceive the stress position in *papel* and *doutor* to be unmarked — just like the stress position in *futuro*, *camada*, and *solene* — a theme vowel consisting solely of one mora without segmental quality, namely *-0* **—** treated as having the same prosodic status as the theme vowels *-o*, *-a*, *-e* in *futur-**o***, *camad-**a***, *solen-**e***, referred to in Portuguese linguistics as *índices temáticos* **—** was postulated, and *papel* and *doutor* were thereby allowed to be analysed as *papel-**0*** and doutor-***0***. On this basis, all of these words were assumed to receive their primary stress through binary foot parsing as in fu(ˈtu.r-**o**), ca(ˈma.d-**a**), so(ˈle.n-**e**), pa(ˈpe.l-**0**), dou(ˈto.r-**0**). In contrast, words such as *câmar-**a*** and *túnel-**0*** were assumed to contain the morphs *camar-* and *tunel-*, which behave as lexical exceptions in stress assignment by requiring that their final syllable initiate binary foot parsing when adjacent to an *índice temático -o*, *-a*, *-e*, or *-0*. Consequently, stress is assigned through prosodification as (ˈcâ.ma)r-**a** and (ˈtú.ne)l-**0**.

Accordingly, since *tele-* and *euro* each constitute a single morpheme — with neither **tel-* nor **eur-* bearing autonomous meaning — the prosodic word structures of *tele-* and *euro* are to be understood as (ˈcâ.ma)r-**a** and (ˈtú.ne)l-**0**, respectively, and are thus prosodified as (ˈte.le)-**0** and (ˈeu.ro)-**0**. The corresponding underlying representations are /tɛl$\overset{\mu}{\text{ɛ}}$-$\overset{\mu}{\ }$/ and /eur$\overset{\mu}{\text{ɔ}}$-$\overset{\mu}{\ }$/, in which the radical final vowels /$\overset{\mu}{\text{ɛ}}$/ and /$\overset{\mu}{\text{ɔ}}$/ become associated with the floating mora of the *índice temático -0* to form derived long vowels $\overset{\mu\mu}{\text{ɛ}}$ and $\overset{\mu\mu}{\text{ɔ}}$. The surface vowels [ɛ] and [ɔ] of *tele-* and *euro* thus appear to reflect these derived long vowels.

(ii) Derived nouns and adjectives formed by suffixation directly to the verbal radical without an intervening thematic vowel.

In Portuguese, one common method of deriving nouns and adjectives from verbs involves the addition of suffixes such as *-ção*, *-dor*, *-tiv-o*, and *-tóri-o* to the verbal radical. In this derivational process, a theme (or thematic) vowel is typically inserted between the verbal radical and the suffix (e.g. [<u>*compar*</u>*-**a***]*r* → [<u>*compar*</u>*-a*]*ção*; [<u>*bat*</u>*-e*]*r* → [<u>*bat*</u>*-e*]*dor*; [<u>*part*</u>*-i*]*r* → [<u>*part*</u>*-i*]*tiv-o*; [<u>*fal*</u>*-a*]*r* → [<u>*fal*</u>*-a*]*tóri-o* — square brackets indicate the verbal stem, and underlining marks the verbal radical). In these derived nouns and adjectives, word stress falls on the suffix — *-ção*, *-dor*, *-tiv-*, *-tóri-* — rendering the radical vowel unstressed. When this vowel is *a*, it undergoes quality reduction — e.g. [*comp*[ɐ]*r-a*]*ção*; [*b*[ɐ]*t-e*]*dor*; [*p*[ɐ]*rt-i*]*tiv-o*; [*f*[ɐ]*l-a*]*tóri-o*.

In contrast to this type of word formation, when such suffixes are attached directly to the verbal base without an intervening thematic vowel, they take the forms *-ão* (underlyingly *-ion*), *-or*, *-iv-o*, *-óri-o*, and the radical vowel *a* often does **not** undergo reduction as in (116):

(116) [*dil*[**ɐ**]*t*-***a***]*r* → [*dil*[**ɐ**]*t*-***a***]*ção*, [*dil*[**ɐ**]*t*-***a***]*dor* ⇔ [*dil*[**a**]*ç*]*ão*, [*dil*[**a**]t]*óri-o*

[*per-su*[**ɐ**]*d*-***i***]*r* → [*per-su*[**ɐ**]*d*-***i***]*ção* ⇔ [*per-su*[**a**]*s*]*ão*, [*per-su*[**a**]*s*]*iv-o*, [*per-su*[**a**]*s*]*óri-o*

[*in-f*[**i**]*r*-***i***]*r* → [*i-l*[**a**]*ç*]*ão*, [*i-l*[**a**]*t*]*iv-o*

The last example involves a suppletive root alternation between *fer-* and *lat-*, but is included here because the lack of vowel reduction appears to correlate with the morphological structure of the word. Barbosa (1988: 353) makes an observation of particular interest in this regard, noting that "... the use of [a] has extended to words like *dissuasão* and *persuasão*, where it alternates with the standard [ɐ]...". This suggests that **the non-reduction of the root vowel *a* in the morphological context [verbal base + non-thematic vowel] + [suffix *-ão, -iv-, -or, -tóri-*]** is **a phenomenon currently in progress from a diachronic perspective**.

It remains unclear whether this phenomenon results from analogy with some existing paradigm. From a morphophonological standpoint, however, it is possible to account for the data by assuming that the stem vowel *-a*, even when not reduced, is underlyingly a short vowel /$\overset{\mu}{\text{a}}$/. On this basis, in surface forms that appear to lack a stem vowel, one may posit, as shown in (117), a stem vowel *-0* with no segmental quality but with a vocalic quantity of one mora — just like the other stem vowels *-a*, *-e*, *-i* — thereby rendering the phenomenon theoretically tractable:

(117) [*dil*$\overset{\mu}{\boldsymbol{a}}$*t*-$\overset{\mu}{\boldsymbol{a}}$]*r* → [*dil*$\overset{\mu}{\boldsymbol{a}}$*t*-$\overset{\mu}{\boldsymbol{a}}$]*ção*; [*dil*$\overset{\mu}{\boldsymbol{a}}$*t*-$\overset{\mu}{\boldsymbol{a}}$]*r* → [*dil*$\overset{\mu}{\boldsymbol{a}}$*ç*-$\overset{\mu}{\boldsymbol{0}}$]*ão*[11]

Then, in forms with the stem vowel *-0*, we may assume that the radical vowel /a/ is relinked to the floating mora once associated with *-0*, yielding a derived long vowel /$\overset{\mu\mu}{\text{a}}$/. This /$\overset{\mu\mu}{\text{a}}$/, like an underlying long vowel /$\overset{\mu\mu}{\text{a}}$/, resists quality reduction even in unstressed syllables as shown in (116):

(118) *dil*$\overset{\mu}{\boldsymbol{a}}$*ç*- $\overset{\mu}{}$**-***ão* → *dil*$\overset{\mu\mu}{\boldsymbol{a}}$*ç*- **-***ão* → *dil*[**a**]*ç*- **-***ão*

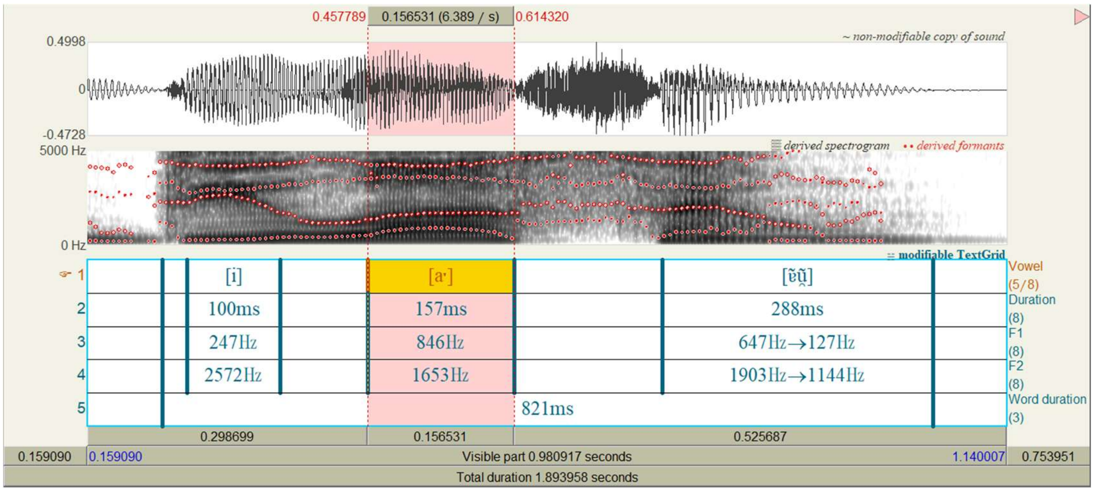
The following are spectrographic analysis images of the audio files available for hearing on *Infopédia*, referenced in footnote 9. Figure (119) corresponds to *dil**a**tação*, and Figure (120) to *dil**a**ção*. The analysis was conducted using Praat (version 64.1.7). As far as we can tell, it is the same native female speaker as (14), (15), (18) and (19). The overall word durations are 821ms for *dil*[aˑ]*ção* and 923ms for *dil*[ɐ]*tação*, probably due to the overall syllable counts. The acoustic measurements of the morphologically same radical vowel *-a-* in *dil**a**ção* and in

[11] As of 5 June 2025, the pronunciations of *dilatação* and *dilação* can be accessed for hearing via the *Infopédia* online dictionary at the following entries:
https://www.infopedia.pt/dicionarios/lingua-portuguesa/dilata%C3%A7%C3%A3o
https://www.infopedia.pt/dicionarios/lingua-portuguesa/dila%C3%A7%C3%A3o

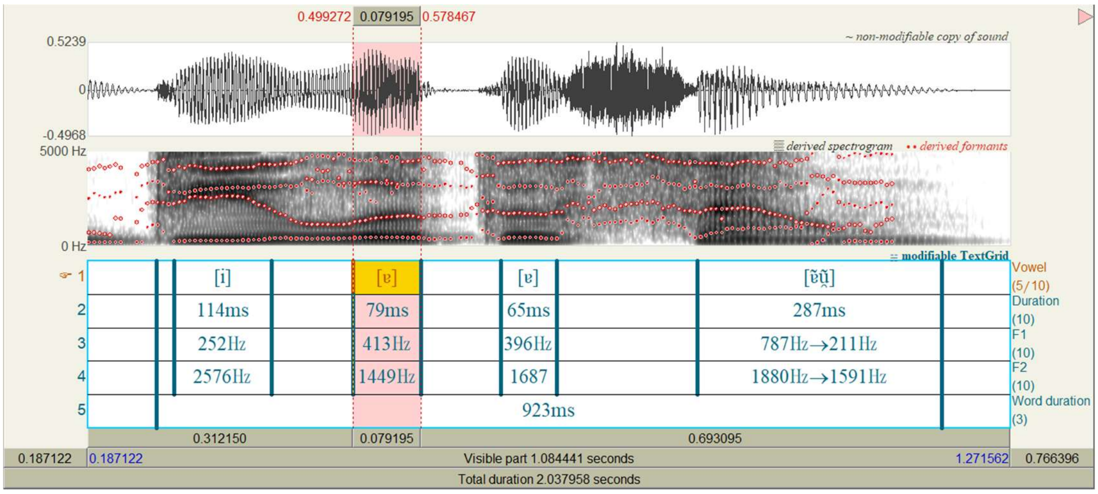
*dil**a**tação* are given in (121) and (122), respectively.

In (119), the radical vowel [aˑ], which is assumed to have two moras, in *dil*[aˑ]*ção* exhibits double the duration (157ms) and a more open quality (F1 = 846 Hz) compared to the morphologically identical vowel [ɐ], which is assumed to have only one mora, in *dil*[ɐ]*tação* (79ms, F1 = 413 Hz) in (120). These examples also, like *p*[aː]*daria* in (14) and *C*[ɐ]*tarina* in (15), are likely to provide phonetic evidence that the qualitatively non-reduced unstressed vowel [a] may exhibit longer duration than the qualitatively reduced unstressed vowel [ɐ] in Contemporary EP, reflecting the difference in underlying quantity.

(119) *dil**a**ção* [dil**a**ˑˈsɐ̃ũ̯]



(120) *dil**a**tação* [dil**ɐ**tɐˈsɐ̃ũ̯]



(121)	Syllable	Vowel	Duration (ms)	F1 (Hz)	F2 (Hz)
	-la-	[aː]	157	846	1653

(122)	Syllable	Vowel	Duration (ms)	F1 (Hz)	F2 (Hz)
	-la-	[ɐ]	79	413	1449

The validity of this morpho-phonological interpretation remains open to further investigation, but it seems clear that forms in which **the stem vowel *a* resists reduction tend to lack a thematic vowel**, and that **this absence is closely related to the lack of reduction**.

(iii) Metrical foot structure and vowel reduction

The present study, as outlined above, has accounted for vowel reduction and non-reduction by means of the underlying or derived mora count of vowels, the presence or absence of primary stress in syllables headed by those vowels, and the theoretical apparatus of constraint hierarchies operating within the Vowel Quality Modification Tier and the Vowel Quantity Adjustment Tier. However, there still remain phenomena that cannot be explained by these factors alone. For instance, *ecónom-o* "manager; steward", *económ-ico* "economic", and *econom-ia* "economy", which all share the word constituent *econom-*, exhibit the following vowel alternations, as shown in (123):

(123) *ec*[ɔ́ː]*n*[u]*m-o* ≈ *ec*[u]*n*[ɔ́ː]*m-ico* ≈ *ec*[ɔ]*n*[u]*m-ia*

Judging from the alternation *ec*[ɔ́ː]*n*[u]*m-o* ≈ *ec*[u]*n*[ɔ́]*m-ico*, the underlying form of the radical in these words appears to be /ik$\overset{\mu}{\text{ɔ}}$nɔm-/, where the first /$\overset{\mu}{\text{ɔ}}$/ is, like the second /ɔ/, an underlying short vowel. In *ec*[ɔ]*n*[u]*m-ia*, nevertheless, this underlying short vowel /$\overset{\mu}{\text{ɔ}}$/ is realised as [ɔ], showing no reduction. One possible explanation lies in the difference in prosodic structure, as illustrated in (124):

(124) (ˌi.k$\overset{\mu}{\text{ɔ}}$)(ˈnɔ.mi)ko ≈ i(ˌk$\overset{\mu}{\text{ɔ}}$.nɔ)(ˈmi.a)

In *ec**o**nómico*, where the underlying /$\overset{\mu}{\text{ɔ}}$/ is realised as reduced [u], the vowel occurs in the complement syllable in the foot (ˌi.k$\overset{\mu}{\text{ɔ}}$). In contrast, in *ec**o**nomia*, where the same underlying /$\overset{\mu}{\text{ɔ}}$/ is realised as unreduced [ɔ], it occurs in the head syllable of the foot (ˌk$\overset{\mu}{\text{ɔ}}$.nɔ). This prosodic difference appears to be related to whether vowel reduction takes place, suggesting that further investigation and research on this point is required.

(iv) Derived words with root vowels in hiatus with a preceding vowel

The root vowels /$\overset{\mu}{\text{e}}$/ and /$\overset{\mu}{\text{ɛ}}$/, when placed in vowel sequences such as *-ae-, -ie-, -oe-, -ue-* — e.g. in *baetilha*, *ajaezar*, *alienar*, *enviesar*, *admoestar*, *amoedar*, *duetista*, and *duelista* — generally do not undergo reduction in contemporary EP, even when they do not bear stress in derivation or verbal inflection, as shown in (125):

(125) *b*[ɐˈeː]*t-a* ≈ *b*[ɐ.e]*t-ilh-a*; *j*[ɐˈeː]*z* ≈ *a-j*[ɐ.e]*z-a-r*; *al*[iˈeː]*n-a* ≈ *al*[i.e]*n-a-r*;

v[iˈɛː]*s* ≈ *en-v*[i.ɛ]*s-a-r*; *adm*[uˈɛː]*st-a* ≈ *adm*[u.ɛ]*st-a-r*; *m*[uˈɛː]*d-a* ≈ *a-m*[u.ɛ]*d-a-r*;

d[uˈeː]*t-o* ≈ *d*[u.e]*t-ist-a*, *d*[u.e]*t-íst-ic-o*; *d*[uˈɛː]*l-o* ≈ *d*[u.ɛ]*l-ist-a*

On the other hand, in Fonseca 1957, many of these examples are transcribed with reduced vowels such as [i] or [ɨ] — e.g. *a-j*[ɐ.i]*z-ar*, *b*[ɐ.ɨ]*t-ilha*, *adm*[u.ɨ]*st-ar*, *a-m*[u.i]*d-ar*, *d*[u.i]*l-ita* — suggesting that the phenomenon whereby unstressed vowels remain unreduced in vowel hiatus contexts is likely to be in progress diachronically. Furthermore, the vowel of the inchoative suffixes *-ec-* and *-esc-* consistently shows reduction after more open vowels in all sources — e.g. *esm*[ɐˈɛː]*ce* ≈ *esm*[ɐ.i]*cer*, *esv*[ɐˈɛː]*ce* ≈ *esv*[ɐ.i]*cer* — but not after more closed vowels, as in *aqu*[iˈɛː]*sce* ≈ *aqu*[i.ɛ]*scer*. This suggests that it is not hiatus alone that determines whether a vowel undergoes reduction, but also whether it belongs to a root or an affix, as well as the aperture relation between the preceding and following vowels. This is therefore a case that calls for further detailed investigation.

7. Conclusion

Although the concept of mora has not traditionally been applied to the analysis of Portuguese vowels, the present study adopts this notion and posits an opposition between the underlying short vowels /$\overset{\mu}{\text{i}}$ $\overset{\mu}{\text{e}}$ $\overset{\mu}{\text{ɛ}}$ $\overset{\mu}{\text{a}}$ $\overset{\mu}{\text{ɔ}}$ $\overset{\mu}{\text{o}}$ $\overset{\mu}{\text{u}}$/ and the underlying long vowels/$\overset{\mu\mu}{\text{ɛ}}$ $\overset{\mu\mu}{\text{a}}$ $\overset{\mu\mu}{\text{ɔ}}$/ in Contemporary European Portuguese. Furthermore, by establishing a system of vowel quality reduction and vowel quantity adjustment arising from the constraint hierarchies operating in two postlexical strata — Postlexical Stratum 1 and Postlexical Stratum 2 — we attempted to account for two types of vowel alternation — [íː éː ɛ́ː áː ɔ́ː óː úː] ≈ [i ɨ ɨ ɐ u u u] and [ɛ́ː áː ɔ́ː] ≈ [ɛ a ɔ] — observed at the phonetic level. Building on this descriptive account, we showed that it is also possible to provide a unified explanation for a range of apparently unrelated phonetic phenomena: the absence of vowel quality reduction in nasal vowels in unstressed syllables; the characteristic vowel alternations in open syllables preceding prepalatal and palatal consonants [ʃ ʒ ʎ ɲ]; the phonetic realisation of vowel coalescence at the level of phrasal phonology; the opposition [ɐ́ː] ≈ [áː] in the verb endings *-amos* and *-ámos* in the first-person plural of the present and preterite indicatives; vowel opening in radical vowels found in derived nouns and adjectives formed by suffixation to verbal radicals lacking intervening thematic vowels; and the absence of vowel reduction in unstressed syllables of derived words with negative implicatures.

Finally, in traditional accounts — whether explicitly or implicitly — it has been assumed that whether an unstressed vowel in a given word undergoes reduction or not is synchronically unpredictable, and must therefore be memorised individually at the lexical level. In other words, speakers are assumed to store not only the "underlying segmental sequence" of each word, but also an additional piece of information regarding whether vowel reduction applies or not. This results in a high degree of lexical redundancy. By contrast, under the present analysis, it suffices in most cases for the speaker to memorise only the underlying segmental sequence of a word at the lexical level. The presence or absence of vowel reduction becomes predictable from the evaluation of output candidates by the constraint rankings operating in each of the lexical and postlexical strata of phonological derivation. As a result, in most cases, the only information that must be stored is the phoneme sequence along with vowel length. This yields a representation that is more compact and principled than in traditional theories, which rely on rote memorisation of exceptional reduction behaviour. At the same time, however, the existence of alternation patterns such as *ec*[ɔ́ː]*n*[u]*m-o* ≈ *ec*[u]*n*[ɔ́ː]*m-ico* ≈ *ec*[ɔ]*n*[u]*m-ia* highlights the need for further investigation and research, as well as for refinement of the theoretical framework.

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